Computing Hilbert modular forms with nontrivial nebentypus

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Abstract. We describe algorithms for computing matrices for the Hecke action on spaces of Hilbert modular forms with nontrivial nebentypus. We also provide self-contained proofs of many statements commonly used in such algorithms which do not have convenient references in the literature.

1. Introduction

Let F be a totally real field of degree n, with ring of integers $\mathbb{Z}_F$. For our purposes, a space of Hilbert modular forms over F is specified by a level $\mathfrak{N} \subset \mathbb{Z}_F$, a weight $k \in \mathbb{Z}^n_{\geq 1}$, and a nebentypus Dirichlet character χ_0 of modulus $\mathfrak{N}$ – we denote this space by $M_k(\mathfrak{N}, \chi_0)$. We write $S_k(\mathfrak{N}, \chi_0)$ for the subspace of cusp forms in $M_k(\mathfrak{N}, \chi_0)$. For any prime $\mathfrak{p} \subset \mathbb{Z}_F$, there is an action of the Hecke operator $T_\mathfrak{p}$ (resp. the diamond operator $\langle \mathfrak{p} \rangle$) on $M_k(\mathfrak{N}, \chi_0)$ which restricts to an action on $S_k(\mathfrak{N}, \chi_0)$. The Jacquet-Langlands correspondence [JL70] gives a Hecke module isomorphism between $S_k(\mathfrak{N}, \chi_0)$ and a space of quaternionic modular forms $S^B_k(\mathfrak{N}, \chi_0)$ for an appropriately chosen quaternion algebra B/F. As such, we can compute matrices for the Hecke action on $S_k(\mathfrak{N}, \chi_0)$ by instead computing matrices for the Hecke action on $S^B_k(\mathfrak{N}, \chi_0)$. Building on work of Eichler [Eic55, ?], Shimura [SHI59], Pizer [Piz80], and others, algorithms for producing matrices for the $T_\mathfrak{p}$-action when F has narrow class number 1, $k = (2, \ldots, 2)$ (parallel weight two), and $\chi_0 = 1$ (trivial nebentypus) were invented and implemented by Greenberg-Voight [GV11] for indefinite B and by Dembélé [Dem07] for definite B. These methods were extended to fields of arbitrary narrow class number and general paritious weights by Voight [Voi10] and Dembélé-Donnelly [DD08] respectively. In this note, we describe algorithms for computing matrices for the Hecke action on spaces of Hilbert modular forms with paritious weight and nontrivial nebentypus.

Main Theorem 1. *Given a totally real field F of degree n, a prime ideal $\mathfrak{p} \subset \mathbb{Z}_F$, a quaternion algebra B/F ramified at all or all but one of the infinite places of F, an order $\mathcal{O} = \mathcal{O}_0(\mathfrak{N}) \subset B$, a paritious weight $k \in \mathbb{Z}^n_{\geq 2}$, and a finitely ramified semigroup homomorphism $\Xi \colon \widehat{\mathcal{O}} \cap \widehat{B}^\times \to \mathbb{C}$ that is trivial on $\mathcal{O}_v \cap B_v^\times$ for $v | \Delta_B$, there is an algorithm that produces a matrix for the action of the Hecke operator $T_\mathfrak{p}$ (resp. the diamond operator $\langle p \rangle$) on $S^B_k(\mathcal{O}, \Xi)$, in a basis independent of $\mathfrak{p}$.*

By the Jacquet-Langlands correspondence, we can use Main Theorem 1 to produce matrices for the Hecke action on $S_k(\mathfrak{N}, \chi_0)$. There is a decomposition

$$S_k(\mathfrak{N}, \chi_0) \cong \bigoplus_\chi S_k(\mathfrak{N}, \chi_0, \chi),$$

where χ ranges over finite-order Hecke characters extending χ_0 (Equation (2)). Note that $S_k(\mathfrak{N}, \chi_0, \chi)$ is often written as $S_k(\mathfrak{N}, \chi)$ (e.g. in [ABB+26]), but we write χ_0 and χ separately to remain consistent with our notation. By considering the action of diamond operators $\{\langle \mathfrak{p} \rangle\}$ for $\mathfrak{p}$ ranging over a set of generators for the narrow class group of F with conductor $\mathfrak{N}\infty$, we can decompose $S_k(\mathfrak{N}, \chi_0)$ into eigenspaces for this diamond action, and obtain the following.

Main Theorem 2. *Let F be a totally real field. Given an integral ideal $\mathfrak{N}$ of F, a paritious weight $k \in \mathbb{Z}^{[F:\mathbb{Q}]}_{\geq 1}$, and a finite order Hecke character χ of F, there is an algorithm that computes a basis of q-expansions (to any given precision) spanning $S_k(\mathfrak{N}, \chi_0, \chi)$.*

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1.1. **Outline.** At a high level, this paper consists of proving Hecke module isomorphisms between various spaces, beginning with definitions coming from the theory of automorphic forms and working towards spaces on which the Hecke action can be effectively computed. We start with the classical definition of the space of automorphic forms for $B^\times$ of weight k, level $\mathcal{O}$, nebentypus Ξ, and central character ω (Equation (1)) – denoted $M_k^B(\mathcal{O}, \Xi, \omega)_{\text{aut}}$ – and the definition of the Hecke action on it. Note that these are $\mathbb{C}$-valued functions. In Section 4 we exhibit a Hecke module isomorphism between $M_k^B(\mathcal{O}, \Xi)_{\text{aut}}$ and the space $M_k^B(\mathcal{O}, \Xi)$ of quaternionic modular forms as they are "usually" defined (Definition 4.1, Equation (8), Equation (9)) – as functions valued in some representation depending on k and Ξ. In Section 5, we give a decomposition of the space of quaternionic modular forms in terms of the connected components of the double coset $B_+^\times \backslash \widehat{B}^\times / \widehat{\mathcal{O}}^\times$ (Lemma 5.2, Lemma 5.3). The ideas of this section, and in particular, the elements $\{\varpi_{j,h}\} \subset B^\times$ defined in Lemma 5.3 play a crucial role in the subsequent sections. In Section 6, we show that the spaces defined in Lemma 5.2 can be identified within group cohomology (Theorem 6.3) using a version of the Eichler-Shimura isomorphism that applies for general Fuchsian groups and general nebentypus character. In doing so, we give a short self-contained proof of the classical Eichler-Shimura isomorphism, relying only on Hodge theory. We then equip these group cohomology spaces with Hecke actions so that the resulting map is a Hecke module isomorphism (Theorem 6.9). Lastly, in Section 7, we generalize the method of Brandt matrices to the setting of nontrivial nebentypus.

For the most part, our algorithms, isomorphisms, and proofs will likely be unsurprising to the experts, and in many cases they are natural generalizations of the corresponding results for classical modular forms. Nevertheless, we felt that it would be beneficial to provide self-contained proofs of these generalizations, as it is difficult to find some of these isomorphisms and their proofs – even in the setting of trivial nebentypus – in the literature. We also hope that this article will be a useful resource for people who are just beginning to study the computation of Hilbert modular forms. We encourage such readers to assume that $\Xi = \omega = 1$ and $h^+(F) = 1$ for a first reading.

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2. Symbols and notation

- $[n]$ – $\{1, \ldots, n\}$;
- F – totally real field with degree n;
- $\mathbb{Z}_F$ – ring of integers of F;
- k – a weight in $\mathbb{Z}^n_{\geq 2}$;
- $\mathcal{H}$ – complex upper half-plane;
- B – quaternion algebra with center F;
- $B_\infty^\times$ – $\prod_{v \mid \infty} B_v^\times$;
- $\mathcal{O}_0(1)$ – maximal order in B;
- $\widehat{B}^\times$ – $\prod'_{v \nmid \infty} (B_v^\times; \mathcal{O}_0(1)_v^\times)$;
- $\mathcal{O}$ – (Eichler) order in B;
- $\widehat{\mathcal{O}}^\times$ – $\prod_{v \nmid \infty} \mathcal{O}_v^\times$;
- Ξ – continuous semigroup homomorphism from $\widehat{\mathcal{O}} \cap \widehat{B}^\times$ to $\mathbb{C}^\times$;
- H – number of right ideal classes in $\mathcal{O}$;
- H_1 – number of right ideal classes of $\mathcal{O}_0(1)$;

Let F^{gal} denote the Galois closure of F over $\mathbb{Q}$. We fix once and for all an embedding $\iota\colon \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$. In particular, this restricts to an embedding $\iota\colon F^{\text{gal}} \hookrightarrow \mathbb{R}$. We also fix an ordering $(\sigma_i)_{i \in [n]}$ of the n embeddings of F into F^{gal}. Given an element $x \in F$, we write $x_i := \sigma_i(x) \in F^{\text{gal}}$. Similarly, for a matrix $\gamma \in M_2(F)$, we write $\gamma_i \in M_2(F^{\text{gal}})$ for the matrix obtained by applying σ_i entrywise.

Given a totally positive element $x \in F^{\text{gal}}$ we define $x^{1/s}$ to be the unique s^{th} root y of x in $\overline{\mathbb{Q}}$ such that $\iota(y) \in \mathbb{R}_{>0}$. We can extend this to define $x^{r/s}$ for any totally positive $x \in F^{\text{gal}}$ and $\frac{r}{s} \in \mathbb{Q}$. We will make frequent use of multi-index notation. For $t \in \mathbb{Q}^n$ and $x \in F^{\text{gal}}$, we write $x^t := \prod_i x_i^{t_i} = \prod_i \sigma_i(x)^{t_i} \in \overline{\mathbb{Q}}$. Similarly, for $z \in \mathbb{C}^n$ and $t \in \mathbb{Z}^n$, we write $z^k := \prod_i z_i^{t_i}$. For $t \in \mathbb{Q}$ and $x \in F^{\text{gal}}$, we write $x^{\underline{t}} := \prod_i x_i^t$.

Throughout, we recall the notions and notations for Hecke characters and for Hilbert modular forms from [ABB+26, Section 2.2]. One distinction, however, is that if χ is a finite order Hecke character of modulus $\mathfrak{N}$ and χ_0 is the Dirichlet character corresponding to χ in the language of [ABB+26], we write $S_k(\mathfrak{N}, \chi_0, \chi)$ for the space of forms which in [ABB+26] is denoted $S_k(\mathfrak{N}, \chi)$.

3. Quaternionic modular forms as automorphic forms

Let B/F be a quaternion algebra split at the first r infinite places and ramified at the last $s := n - r$ infinite places (we can always reorder the places of F so that this holds). Fix a maximal compact subgroup K_∞ of $B_\infty^\times := \prod_{v|\infty} B_v^\times \cong (\mathrm{GL}_2(\mathbb{R}))^r \times (\mathbb{H}^\times)^s$. Fix a maximal order $\mathcal{O}_0(1) \subset B$, and at each finite place v of F, define a hyperspecial compact subgroup $K_v := \mathcal{O}_0(1)_v^\times \subset B_v^\times$. With these choices in hand, we may define the group $B^\times(\mathbb{A}_F) := \prod'_v(B_v^\times; K_v)$. Let $\widehat{K} := \prod_{v\nmid\infty} K_v$ and $K := K_\infty \times \widehat{K}$. where $B^\times$ embeds into $B^\times(\mathbb{A}_F)$ diagonally. An automorphic form for $B^\times(\mathbb{A}_F)$ with central character $\omega\colon \mathbb{A}_F^\times \to S^1$ is a function

$$\varphi\colon B^\times\backslash B^\times(\mathbb{A}_F) \longrightarrow \mathbb{C}$$

that is K-finite (the span of right translates of φ by $k \in K$ is finite-dimensional), has central character ω, and satisfies a moderate growth condition. Fix an order $\mathcal{O} \subset \mathcal{O}_0(1)$ and a continuous semigroup homomorphism $\Xi\colon \widehat{\mathcal{O}} \cap \widehat{B}^\times \to \mathbb{C}$ such that $\Xi|_{\mathbb{A}_F^\times \cap \widehat{\mathcal{O}}} = \omega|_{\mathbb{A}_F^\times \cap \widehat{\mathcal{O}}}$.

If the local semigroup homomorphism $\Xi_\mathfrak{p}\colon \mathcal{O}_\mathfrak{p} \cap B_\mathfrak{p}^\times \to \mathbb{C}$ is nontrivial at finitely many finite places of F, we say that Ξ is finitely ramified. For every such $\mathfrak{p}$, by continuity, there exists a minimal $f_\mathfrak{p}$ such then $\Xi_\mathfrak{p}|_{\mathcal{O}_\mathfrak{p}^\times}$ factors through $\mathcal{O}_\mathfrak{p}^\times/(1 + \mathfrak{p}^{f_\mathfrak{p}}\mathcal{O}_\mathfrak{p})$ and call $\mathfrak{f} = \prod_\mathfrak{p} \mathfrak{p}^{f_\mathfrak{p}}$ the conductor of Ξ.

Example 3.1. Consider the automorphic forms corresponding to classical modular forms of level N and nebentypus a finite-order Hecke character χ. In this case, $B = M_{2,\mathbb{Q}}$ and $\widehat{\mathcal{O}} \subseteq M_2(\widehat{\mathbb{Z}})$ consists of the matrices which are upper triangular modulo N. Following the notation of [ABB+26, Section 2.2], we write χ_0 for the Dirichlet character obtained by restricting the Hecke character χ to the finite places. Abusing notation, we obtain a semigroup homomorphism

$$\begin{aligned} \chi\colon \quad M_2(\hat{\mathbb{Z}}) &\longrightarrow \mathbb{C} \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} &\longmapsto \chi_0^*(d) \end{aligned},$$

where

$$\chi_0^*(d) := \begin{cases} \chi_0(d) & (d, N) = 1 \\ 0 & \text{otherwise.} \end{cases}$$

In this case, the conductor of χ as a semigroup homomorphism is the same as its conductor as a Hecke character (which is the same as the conductor of χ_0 as a Dirichlet character).

If $\mathcal{O}$ is an Eichler order $\mathcal{O}_0(\mathfrak{N})$ and B is split at every prime dividing $\mathfrak{N}$, any character $\chi_0\colon (\mathbb{Z}_F/\mathfrak{N})^\times \to \mathbb{C}^\times$ corresponds to a semigroup homomorphism on $\widehat{\mathcal{O}_0(\mathfrak{N})}$ exactly as in Example 3.1, which we also denote by χ_0.

We say that an automorphic form φ has level $\mathcal{O}$ and nebentypus Ξ if for any $\hat{u} \in \widehat{\mathcal{O}}^\times$, $\varphi(x\hat{u}) = \Xi(\hat{u})^{-1}\varphi(x)$. The choice of inverse here may appear unnatural, but it is chosen to make the definitions in Section 5 and Section 6 agree with the usual definition of modular form with nebentypus. A quaternionic modular form is an automorphic form for $B^\times$ satisfying an additional condition at the infinite real places. Let

$$B_{\infty,\mathbb{R}}^\times := \prod_{\substack{v|\infty \\ v \text{ split}}} B_v^\times \cong \mathrm{GL}_2(\mathbb{R})^r \qquad \text{and} \qquad B_{\infty,\mathbb{C}}^\times := \prod_{\substack{v|\infty \\ v \text{ ramified}}} B_v^\times \cong (\mathbb{H}^\times)^s.$$

Similarly, let $K_{\infty,\mathbb{R}} := K_\infty \cap B_{\infty,\mathbb{R}} \cong \mathrm{O}_2(\mathbb{R})^r$ and $K_{\infty,\mathbb{C}} := K_\infty \cap B_{\infty,\mathbb{C}} \cong \mathrm{U}_2(\mathbb{R})^s$.

Definition 3.2. A quaternionic modular form for $B^\times$ of weight k, level $\mathcal{O}$, and nebentypus Ξ is a function

$$\varphi\colon B^\times\backslash B^\times(\mathbb{A}_F) \longrightarrow \mathbb{C}$$

such that

(1) For any $\hat{u} \in \widehat{\mathcal{O}}^\times$, $\varphi(x\hat{u}) = \Xi(\hat{u})^{-1}\varphi(x)$;
(2) φ is K-finite;

(3) If v is an infinite place where B is split, the local component at v of the automorphic representation associated to φ is holomorphic discrete series of weight k_v. In particular, for $\kappa_v = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}^{\epsilon} \in K_{\infty,\mathbb{R}}$,

$$\varphi(x\kappa_v) = \frac{|\det \kappa_v|^{k_v/2}}{j(\kappa_v, \sqrt{-1})^{k_v}} \varphi(x) = e^{-(-1)^{\epsilon}\sqrt{-1}k_v\theta}\varphi(x). \tag{1}$$

(4) If v is an infinite place where B is ramified, the local component at v of the automorphic representation associated to φ is isomorphic to $\mathrm{Sym}^{k_v-2}\,\mathbb{C}^2$.

The space of such forms is denoted $M_k^B(\mathcal{O},\Xi)_{\mathrm{aut}}$. The subspace of $M_k^B(\mathcal{O},\Xi)_{\mathrm{aut}}$ with central character ω is denoted $M_k^B(\mathcal{O},\Xi,\omega)_{\mathrm{aut}}$.

Remark. It is necessary to specify Ξ and ω separately. For example, suppose that $\mathcal{O} = \mathcal{O}_0(\mathfrak{N})$. Then, $\mathbb{A}_F^\times \cap \widehat{\mathcal{O}} = \prod_{v\nmid\infty} \mathbb{Z}_{F,v}^\times =: \widehat{\mathbb{Z}}_F^\times$, so Ξ determines ω_0 in the language of [ABB+26, Section 2]. The unitary Hecke characters ω extending $\omega_0 = \Xi^{-1}|_{\widehat{\mathbb{Z}}_F}^{\times}$ form a $(\mathrm{Cl}_F^+)^\vee$-torsor, since we can multiply any such extension ω by a character of Cl_F^+ to get another extension ω' with $\omega_0 = \omega_0'$. For general $\mathcal{O}$, $\mathbb{A}_F^\times \cap \widehat{\mathcal{O}}$ can be even smaller, and the gap between Ξ and ω may be even more pronounced.

As noted earlier, there is a decomposition

$$M_k^B(\mathcal{O},\Xi)_{\mathrm{aut}} \cong \bigoplus_{\omega} M_k^B(\mathcal{O},\Xi,\omega)_{\mathrm{aut}} \tag{2}$$

where ω ranges over finite-order Hecke characters of F extending $\Xi^{-1}|_{\mathbb{A}_F^\times \cap \widehat{\mathcal{O}}}$.

We proceed to define the Hecke action on $M_k^B(\mathcal{O},\Xi,\omega)_{\mathrm{aut}}$. Given an element $\hat{\alpha} \in \widehat{B}^\times$, there exists an $A \in \mathbb{Z}$ and elements $\{\hat{\alpha}_j\}_{j=1}^A \subset \widehat{B}^\times$ such that

$$\widehat{\mathcal{O}}^\times \backslash \widehat{\mathcal{O}}^\times \hat{\alpha} \widehat{\mathcal{O}}^\times = \bigsqcup_{j=1}^{A} \widehat{\mathcal{O}}^\times \hat{\alpha}_j. \tag{3}$$

We define

$$(T_{\hat{\alpha}}\varphi)(x) := \sum_{j=1}^{A} \varphi(x\hat{\alpha}_j^{-1})\Xi(\hat{\alpha}_j)^{-1}. \tag{4}$$

This definition comes from the theory of automorphic representations (see e.g. [?, Chapter 3]). By the first property in Definition 3.2, Equation (4) depends only on the cosets $\{\widehat{\mathcal{O}}^\times \hat{\alpha}_j\}$ and not on the specific representatives $\{\hat{\alpha}_j\}$. Definition 3.2 and Equation (4) together define the space of quaternionic modular forms as a Hecke module.

The Hecke algebra is the algebra generated by all the $T_{\hat{\alpha}}$ for $\hat{\alpha} \in \widehat{\mathcal{O}}^\times \backslash \widehat{B}^\times / \widehat{\mathcal{O}}^\times$. Given a prime ideal $\mathfrak{p} \subset \mathbb{Z}_F$, let $\varpi \in F_{\mathfrak{p}}$ be a uniformizer and $\pi \in B_{\mathfrak{p}}^\times$ an element such that $\mathrm{nrd}(\pi) = \varpi$. We introduce the operators $T_{\mathfrak{p}} = T_{\hat{\pi}}$ (the Hecke operator at $\mathfrak{p}$) and $\langle \mathfrak{p} \rangle = T_{\hat{\varpi}}$ (the diamond operator at $\mathfrak{p}$), where $\hat{\pi}, \hat{\varpi} \in \widehat{B}^\times$ are elements which are trivial at places $v \neq \mathfrak{p}$ and at $v = \mathfrak{p}$ equal π and ϖ, respectively. Equivalently, we may choose $\hat{\pi}, \hat{\varpi}$ such that $\mathrm{nrd}(\hat{\pi})\widehat{\mathbb{Z}}_F \cap F = \hat{\varpi}\widehat{\mathbb{Z}}_F \cap F = \mathfrak{p}$.

As in Equation (3), there exists a $P \in \mathbb{Z}$ and elements $\{\hat{\pi}_j\}_{j=1}^P \subset \widehat{B}^\times$ such that

$$\widehat{\mathcal{O}}^\times \backslash \widehat{\mathcal{O}}^\times \hat{\pi} \widehat{\mathcal{O}}^\times = \bigsqcup_{j=1}^{P} \widehat{\mathcal{O}}^\times \hat{\pi}_j. \tag{5}$$

When $\mathcal{O}$ is $\mathcal{O}_0(\mathfrak{N})$, P is $\mathrm{Nm}(\mathfrak{p})+1$ if $\mathfrak{p} \nmid \mathfrak{N}$ and $\mathrm{Nm}(\mathfrak{p})$ otherwise.

It follows from Equation (4) that

$$(T_{\mathfrak{p}}\varphi)(x) := \sum_{j=1}^{P} \varphi(x\hat{\pi}_j^{-1})\Xi(\hat{\pi}_j)^{-1}. \tag{6}$$

Similarly, as $\varpi \in F_{\mathfrak{p}}$ commutes with $\mathcal{O}_{\mathfrak{p}}$, the double coset decomposition in Equation (3) is trivial, so in this case Equation (4) becomes

$$(\langle \mathfrak{p} \rangle \varphi)(x) := \varphi(x\hat{\varpi}^{-1})\Xi(\hat{\varpi})^{-1}. \tag{7}$$

Let η be a Hecke character of F and consider the function $\varphi_\eta := \eta \circ \mathrm{nrd}$ – it is an automorphic form for $B^\times$. Suppose $\varphi_\eta \in M_k(\mathcal{O}, \Xi\omega)$. By Definition 3.2, $\Xi^{-1}|_{\widehat{\mathcal{O}}^\times} = \varphi_\eta|_{\widehat{\mathcal{O}}^\times}$. The function φ_η has central character η^2, so $\omega = \eta^2$. Because right translates of φ_η are scalar multiples of φ_η, the automorphic representation associated to φ_η is one-dimensional. It follows from Definition 3.2 that B is definite and $k = (\underline{2})$. We write $\mathcal{E}_k^B(\mathcal{O}, \Xi, \omega)$ for the subspace of $M_k^B(\mathcal{O}, \Xi, \omega)$ spanned by functions of the form $\{\varphi_\eta\}$. As we will see in Section 4, for $k = (\underline{2})$ and B definite, $M_k^B(\mathcal{O}, \Xi, \omega)$ consists of complex-valued functions on a finite set. In this case, we can define the standard inner product on such functions, and define $S_k^B(\mathcal{O}, \Xi, \omega)$ to be the orthogonal complement of $\mathcal{E}_k^B(\mathcal{O}, \Xi, \omega)$. Otherwise, if $B \neq M_{2,F}$ and either $k \neq (\underline{2})$ or B is not totally definite, we define $S_k^B(\mathcal{O}, \Xi, \omega) = M_k^B(\mathcal{O}, \Xi, \omega)$. Even though the Shimura variety associated to B has no cusps when $B \neq M_2(F)$, we nonetheless call $S_k^B(\mathcal{O}, \Xi, \omega)$ the space of quaternionic cusp forms[1].

3.1. **The Jacquet-Langlands correspondence and Hilbert modular forms.** Our primary motivation for computing spaces of quaternionic modular forms is that they are isomorphic to certain spaces of Hilbert modular forms. Let B be a quaternion algebra unramified outside the infinite places.

Given an ideal $\mathfrak{D} \subset \mathbb{Z}_F$ and a space of Hilbert cusp forms $S_k(\mathfrak{N}, \Xi)$, the $\mathfrak{D}$-new subspace of $S_k(\mathfrak{N}, \Xi)$ is the complement of the images of the degeneracy maps at $\mathfrak{p}$ for all primes $\mathfrak{p}|\mathfrak{D}$.

Theorem 3.3 (Jacquet-Langlands (see e.g. [Hid81, Theorem 2.12]))**.** *Let $k \in \mathbb{Z}_{\geq 2}^n$ be a weight, $\mathfrak{N} \subset \mathbb{Z}_F$ a level, and χ a finite-order Hecke character. Let B/F be a quaternion algebra with discriminant Δ_B such that the conductor of χ divides $\mathfrak{N}\Delta_B^{-1}$. Then, there is a Hecke module isomorphism*

$$S_k^B(\mathcal{O}_0(\mathfrak{N}\Delta_B^{-1}), \chi) \cong S_k(\mathfrak{N}, \chi)^{\Delta_B\text{-new}}.$$

By Theorem 3.3, the Hecke action on the $\mathfrak{D}$-new subspace of $S_k(\mathfrak{N}, \chi)$ is the same as the Hecke action on $S_k^B(\mathfrak{N}, \chi)$. In particular, if $\Delta_B = (1)$, then $S_k^B(\mathcal{O}_0(\mathfrak{N}), \chi) \cong S_k(\mathfrak{N}, \chi)$ on the nose by Theorem 3.3. Given any subset of places of F of even cardinality, there exists a quaternion algebra B/F ramified at exactly those places [Voi21, Proposition 27.5.15]. When n is even, we can choose B to be a definite quaternion algebra ramified at every infinite place ($r = 0$) and unramified away from the infinite places. When n is odd, we can choose B to be an indefinite quaternion algebra ramified at all but one of the infinite places ($r = 1$) and unramified away from the infinite places. It follows that for any totally real field, we can choose a B for which the Hecke matrices on $S_k^B(\mathcal{O}_0(\mathfrak{N}), \chi)$ are exactly the Hecke matrices on $S_k(\mathfrak{N}, \chi)$.

Because $S_k(\mathfrak{N}, \chi)$ is spanned by Hecke eigenforms, a basis of q-expansions for $S_k(\mathfrak{N}, \chi)$ can be recovered from matrices for the $T_\mathfrak{p}$-action (for sufficiently any primes $\mathfrak{p} \subset \mathbb{Z}_F$) on $S_k^B(\mathfrak{N}, \chi)$. This was made algorithmic by [DV21]. As such, the problem of computing a basis of $S_k(\mathfrak{N}, \chi)$ reduces to computing Hecke matrices on $S_k^B(\mathcal{O}_0(\mathfrak{N}), \chi)$.

Unfortunately, the description of quaternionic modular forms given by Definition 3.2 and Equation (6) is not amenable to explicitly computing Hecke matrices. In what follows, we will deduce from Definition 3.2 and Equation (6) several equivalent descriptions of the space of quaternionic modular forms as a Hecke module, and ultimately give algorithms for computing Hecke matrices on spaces of quaternionic modular forms with nontrivial nebentypus.

Main Theorem 1. *Given a totally real field F of degree n, a prime ideal $\mathfrak{p} \subset \mathbb{Z}_F$, a quaternion algebra B/F ramified at all or all but one of the infinite places of F, an order $\mathcal{O} = \mathcal{O}_0(\mathfrak{N}) \subset B$, a paritious weight $k \in \mathbb{Z}_{\geq 2}^n$, and a finitely ramified semigroup homomorphism $\Xi\colon \widehat{\mathcal{O}} \cap \widehat{B}^\times \to \mathbb{C}$ that is trivial on $\mathcal{O}_v \cap B_v^\times$ for $v|\Delta_B$, there is an algorithm that produces a matrix for the action of the Hecke operator $T_\mathfrak{p}$ (resp. the diamond operator $\langle p \rangle$) on $S_k^B(\mathcal{O}, \Xi)$, in a basis independent of $\mathfrak{p}$.*

The methods in this paper will actually work for more general $\mathcal{O}$, but some of the algorithms in [DV13] which we call as subroutines assume $\mathcal{O}$ to be of this form. We expect this requirement can be relaxed without much difficulty, but do not treat this here. Combining Main Theorem 1 with the algorithms of [DV21] and the Hecke stability method (described in [Sch15] for modular forms and extended in [MS15, ABB$^+$26] to Hilbert modular forms) for computing partial weight 1 forms, we obtain the following.

[1]The forms in $\mathcal{E}_k^B(\mathcal{O}, \Xi, \omega)$ are sometimes called quaternionic Eisenstein series. However, I find this terminology confusing as the representations associated to these forms are not in the continuous spectrum.

Main Theorem 2. *Let F be a totally real field. Given an integral ideal $\mathfrak{N}$ of F, a paritious weight $k \in \mathbb{Z}_{\geq 1}^{[F:\mathbb{Q}]}$, and a finite order Hecke character χ of F, there is an algorithm that computes a basis of q-expansions (to any given precision) spanning $S_k(\mathfrak{N}, \chi_0, \chi)$.*

4. The "usual" definition of quaternionic modular forms

Let $\varphi \in M_k^B(\mathcal{O}, \Xi)_{\mathrm{aut}}$ be a quaternionic modular form as in Definition 3.2. Consider the space W spanned by the right translates of φ by $(\mathbb{H}^\times)^s$. As a Lie group, $\mathbb{H}^\times \cong \mathrm{SU}_2(\mathbb{R}) \times \mathbb{R}_{>0}^\times$ – the former is compact and the latter is in the center. Because φ is K-finite and has a central character, W is a twist of a finite-dimensional irreducible representation of $\mathrm{SU}_2(\mathbb{R})^s$ by a power of the determinant (equivalently, the reduced norm of $\mathbb{H}^\times$) for each factor. As a representation, we can write

$$W \cong \bigotimes_{i=1}^{s} (\mathrm{Sym}^{k_i - 2}\,\mathbb{C}^2) \otimes \det{}^{l_i}$$

for $k_i \in \mathbb{Z}_{\geq 2}$ and $l_i \in \mathbb{C}$ for all $i \in [s]$. Because we require ω to be unitary, we must have $l_i = \frac{2-k_i}{2}$ – the square root makes sense here because the reduced norm of an element of $\mathbb{H}^\times$ is always positive. Because $\mathbb{H}^\times$ acts on functions in W by right translation, W is equipped with a left $\mathbb{H}^\times$-action. We write W_k for the space of functionals $f \colon W \to \mathbb{C}$ equipped with the right action of $h \in \mathbb{H}^\times$ given by $(f \cdot h)(\varphi') := f(h \cdot \varphi')$. In what follows, we assume for ease of notation that B is split at the first r infinite places, and write $k_{\mathbb{R}} := (k_1, \ldots, k_r)$ to denote the components of the weight at these places. We also abbreviate $j(\gamma, z)^{k_{\mathbb{R}}} := \prod_{i=1}^r j(\gamma_i, z_i)^{k_i}$.

The definition of $M_k^B(\mathcal{O}, \Xi)$ that one sees more often (e.g. [DV13, Lemma 7.11]) is the following.

Definition 4.1. A quaternionic modular form for $B^\times$ of weight k, level $\mathcal{O}$, and nebentypus Ξ is a function

$$\phi \colon \mathcal{H}^r \times \widehat{B}^\times \longrightarrow W_k$$

such that

(1) For any $\gamma \in B_+^\times$,
$$\phi(z, \hat{x}) = \frac{(\det \gamma)^{k_{\mathbb{R}}/2}}{j(\gamma, z)^{k_{\mathbb{R}}}} \phi(\gamma z, \gamma \hat{x} \widehat{\mathcal{O}}^\times)^\gamma.$$

(2) For any $\hat{u} \in \widehat{\mathcal{O}}^\times$,
$$\phi(z, \hat{x}\hat{u}) = \Xi(\hat{u})^{-1} \phi(z, \hat{x}).$$

(3) ϕ is holomorphic in the first variable.

The space of such forms is denoted $M_k^B(\mathcal{O}, \Xi)$. The subspace of $M_k^B(\mathcal{O}, \Xi)$ with central character ω is denoted $M_k^B(\mathcal{O}, \Xi, \omega)$.

We can define the Hecke operator $T_{\mathfrak{p}}$ on ϕ as in Definition 4.1 by the formula

$$T_{\mathfrak{p}}\phi(z, \hat{x}) = \sum_{j=1}^{P} \phi(z, \hat{x}\hat{\pi}_j^{-1}) \Xi(\hat{\pi}_j)^{-1}. \tag{8}$$

As with Equation (6), the right-hand side of Equation (8) is independent of the choices of $\{\hat{\pi}_j\}$. Similarly, we can define the diamond operator $\langle p \rangle$ on ϕ by the formula

$$\langle p \rangle \phi(z, \hat{x}) = \phi(z, \hat{x}\hat{\varpi}^{-1}) \Xi(\hat{\varpi})^{-1}. \tag{9}$$

Here is the main result of this section.

Proposition 4.2. *As Hecke modules, $M_k^B(\mathcal{O}, \Xi, \omega) \cong M_k^B(\mathcal{O}, \Xi, \omega)_{\mathrm{aut}}$.*

We will prove Proposition 4.2 by defining spaces $M_k^B(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}}$ and $M_k^B(\mathcal{O}, \Xi, \omega)_{\pm}$, and Hecke module isomorphisms

$$M_k^B(\mathcal{O}, \Xi, \omega)_{\mathrm{aut}} \xrightarrow{\sim} M_k^B(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}} \xrightarrow{\sim} M_k^B(\mathcal{O}, \Xi, \omega)^{\pm} \xrightarrow{\sim} M_k^B(\mathcal{O}, \Xi, \omega).$$

Definition 4.3. A quaternionic modular form of level $\mathcal{O}$ and nebentypus Ξ is a function

$$\tilde{\varphi} \colon B^\times \backslash B^\times(\mathbb{A}_F) \longrightarrow W_k$$

such that

(1) For any $\hat{u} \in \widehat{\mathcal{O}}^\times$, $\tilde{\varphi}(x\hat{u}) = \Xi(\hat{u})\tilde{\varphi}(x)$
(2) For any $\kappa \in K_{\infty,\mathbb{R}}$,
$$\tilde{\varphi}(x\kappa) = \frac{|\det \kappa|^{\frac{k}{2}}}{j(\kappa, \sqrt{-1})^k}\tilde{\varphi}(x). \tag{10}$$
(3) For any $h \in B^\times_{\infty,\mathbb{C}}$, $\tilde{\varphi}(xh) = \tilde{\varphi}(x)^h$.

The space of such forms is denoted $M_k^B(\mathcal{O}, \Xi)'_{\mathrm{aut}}$. The subspace of $M_k^B(\mathcal{O}, \Xi)'_{\mathrm{aut}}$ with central character ω is denoted $M_k^B(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}}$.

We define
$$(T_\mathfrak{p}\tilde{\varphi})(x) := \sum_{j=1}^{P} \tilde{\varphi}(x\hat{\pi}_j^{-1})\Xi(\hat{\pi}_j)^{-1}, \tag{11}$$
and similarly
$$(\langle\mathfrak{p}\rangle\tilde{\varphi})(x) := \tilde{\varphi}(x\hat{\varpi}^{-1})\Xi(\hat{\varpi})^{-1}. \tag{12}$$

Write ${}_h\varphi \in W$ for the right translate of $\varphi \in M_k^B(\mathcal{O}, \Xi, \omega)_{\mathrm{aut}}$ by $h \in B^\times_{\infty,\mathbb{C}} \cong (\mathbb{H}^\times)^s$.

Lemma 4.4. *The map*
$$\begin{aligned} \Phi' \colon\ M_k^B(\mathcal{O}, \Xi, \omega)_{\mathrm{aut}} &\longrightarrow M_k^B(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}} \\ \varphi &\longmapsto (x \longmapsto ({}_h\varphi \longmapsto {}_h\varphi(x))) \end{aligned}$$
is a Hecke module isomorphism.

Proof. The map is well-defined – the first two properties of Definition 4.3 follow because right translation by $B^\times_{\infty,\mathbb{C}}$ commutes with right translation by $K_{\infty,\mathbb{R}}$ and by $\widehat{\mathcal{O}}^\times$. Φ' has an inverse map given by sending $x \in \mathbb{A}_F^\times$ to $(\Phi'(\phi))(\phi)$ (the restriction of the functional $\Phi'(\phi)$ to ϕ). Because right translation by $h \in B^\times_{\infty,\mathbb{C}}$ commutes with right translation by elements of $\widehat{B}^\times$, the map is a Hecke module isomorphism by Equation (11) and Equation (12). ☐

Observe that $M_k^B(\mathcal{O}, \Omega, \omega)_{\mathrm{aut}}$ and $M_k^B(\mathcal{O}, \Omega, \omega)'_{\mathrm{aut}}$ are spaces of right K-equivariant and left $B^\times$-invariant functions. The point of introducing $M_k^B(\mathcal{O}, \Omega, \omega)'_{\mathrm{aut}}$ is that it helps us define a space of right K_∞-invariant functions.

Let $M_k^B(\mathcal{O}, \Xi, \omega)^\pm$ denote the space of functions on $\mathcal{H}^{\pm r} \times \widehat{B}^\times \to W_k$ satisfying the conditions of Definition 4.1 with the first condition replaced by

(1') For any $\gamma \in B^\times$,
$$\phi(z, \hat{x}) = \frac{|\det \gamma|^{k_\mathbb{R}/2}}{j(\gamma, z)^{k_\mathbb{R}}}\phi(\gamma z, \gamma\hat{x}\widehat{\mathcal{O}}^\times)^\gamma.$$

The Hecke action on $M_k^B(\mathcal{O}, \Xi, \omega)^\pm$ is still given by Equation (8).

By weak approximation [Voi21, Proposition 28.7.3(b)], $B^\times$ has elements of all possible signs at the real places. It follows that the map from $M_k^B(\mathcal{O}, \Xi, \omega)^\pm \to M_k^B(\mathcal{O}, \Xi, \omega)$ given by restriction to $\mathcal{H}^r \times \widehat{B}^\times$ is a Hecke module isomorphism. To prove Proposition 4.2 it suffices to prove that $M_k^B(\mathcal{O}, \Omega, \omega)'_{\mathrm{aut}} \cong M_k^B(\mathcal{O}, \Omega, \omega)^\pm$. In what follows, we write $x \in B^\times(\mathbb{A}_F)$ as $(x_\mathbb{R}, x_\mathbb{C}, \hat{x}) \in B^\times_{\infty,\mathbb{R}} \times B^\times_{\infty,\mathbb{C}} \times \widehat{B}^\times \cong \widehat{B}^\times(\mathbb{A}_F)$.

Lemma 4.5. *The map Φ sending $\tilde{\varphi} \in M_k^B(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}}$ to the function*
$$\begin{aligned} \Phi(\tilde{\varphi}) \colon\ B^\times_{\infty,\mathbb{R},+} \times \widehat{B}^\times &\longrightarrow W_k \\ (x_\mathbb{R}, \hat{x}) &\longmapsto \tfrac{j(x_\mathbb{R}, \sqrt{-1})^{k_\mathbb{R}}}{|\det x_\mathbb{R}|^{k_\mathbb{R}/2}}\tilde{\varphi}(1, x_\mathbb{R}, \hat{x}), \end{aligned}$$
induces a Hecke module isomorphism $\Phi \colon M_k(\mathcal{O}, \Xi, \omega)'_{\mathrm{aut}} \to M_k(\mathcal{O}, \Xi, \omega)^\pm$.

Proof. The function $\Phi(\tilde{\varphi})$ is right $K_{\infty,\mathbb{R}}$-invariant by Equation (10). As such, we can think of ϕ as a function on
$$B^\times_{\infty,\mathbb{R}}/K_{\infty,\mathbb{R}} \times \widehat{B}^\times \cong \mathcal{H}^{\pm r} \times \widehat{B}^\times.$$

Furthermore,

$$\frac{|\det\gamma_{\mathbb{R}}|^{k_{\mathbb{R}}/2}}{j(\gamma_{\mathbb{R}},z)^{k_{\mathbb{R}}}}\phi(\gamma_{\mathbb{R}}x_{\mathbb{R}},\hat{\gamma}\hat{x})^{\gamma_{\mathbb{C}}} = \frac{|\det\gamma_{\mathbb{R}}|^{k_{\mathbb{R}}/2}}{j(\gamma_{\mathbb{R}},z)^{k_{\mathbb{R}}}}\frac{j(\gamma_{\mathbb{R}}x_{\mathbb{R}},\sqrt{-1})^{k_{\mathbb{R}}}}{|\det\gamma_{\mathbb{R}}x_{\mathbb{R}}|^{k/2}}\tilde{\varphi}(\gamma(\gamma_{\mathbb{C}}^{-1},x_{\mathbb{R}},\hat{x}))^{\gamma_{\mathbb{C}}} = \frac{j(x_{\mathbb{R}},\sqrt{-1})^{k_{\mathbb{R}}}}{|\det x_{\mathbb{R}}|^{k_{\mathbb{R}}/2}}\tilde{\varphi}(1,x_{\mathbb{R}},\hat{x}) = \phi(x_{\mathbb{R}},\hat{x})$$

and

$$\phi(x_{\mathbb{R}},\hat{x}\hat{u}) = \tilde{\varphi}(1,x_{\mathbb{R}},\hat{x}\hat{u}) = \Xi(\hat{u})^{-1}\phi(x_{\mathbb{R}},\hat{x}).$$

Therefore, $\phi\in M_k^B(\mathcal{O},\Xi,\omega)^{\pm}$. Similarly, the map Ψ sending $\phi\in M_k^B(\mathcal{O},\Xi,\omega)^{\pm}$ to

$$\Psi(\phi)(x_{\mathbb{C}},x_{\mathbb{R}},\hat{x}) := \frac{|\det x_{\mathbb{R}}|^{\frac{k}{2}}}{j(x_{\mathbb{R}},\sqrt{-1})^{k}}\phi(x_{\mathbb{R}},\hat{x})^{x_{\mathbb{C}}}$$

has image in $M_k^B(\mathcal{O},\Xi,\omega)'_{\mathrm{aut}}$, and we can check that it is the inverse of Φ. The map Φ is compatible with the Hecke actions in Equation (6) and Equation (8), so it is a Hecke module isomorphism. □

5. Preliminaries on computing quaternionic modular forms with nebentypus

One issue with the description of Definition 4.1 is that it is not immediately clear how to produce an explicit basis for the space. Even given a basis, the formula of Equation (8) involves multiplication in the adèles, which is computationally inconvenient. In this section, we will produce a decomposition of $S_k^B(\mathcal{O},\Xi)$ (Lemma 5.2) and an alternative description of the Hecke action on $S_k^B(\mathcal{O},\Xi)$ in terms of this decomposition (Lemma 5.3). In particular, the formula of Lemma 5.3 will be entirely in terms of elements of $B^\times$, rather than elements of $B^\times(\mathbb{A}_F)$.

5.1. **Components of the quaternionic Shimura variety.** While the forms in $M_k^B(\mathcal{O},\Xi)$ (as in Definition 4.1) are not quite right $\widehat{\mathcal{O}}^\times$-invariant when Ξ is nontrivial, they are nonetheless determined by their values on representatives in

$$Y^B(\mathcal{O}) := B_+^\times\backslash\left(\mathcal{H}^r\times\widehat{B}^\times/\widehat{\mathcal{O}}^\times\right). \tag{13}$$

We call $Y^B(\mathcal{O})$ the quaternionic Shimura variety associated to $\mathcal{O}$.

Remark. When $B\not\cong M_{2,F}$, $Y^B(\mathcal{O})$ has no cusps, and $Y^B(\mathcal{O}) = X_B(\mathcal{O})$ is compact.

Because $\mathcal{H}^r$ is connected, the connected components of $Y^B(\mathcal{O})$ correspond to the elements of $B_+^\times\backslash\widehat{B}^\times/\widehat{\mathcal{O}}^\times$.

Theorem 5.1 ([Voi21, 28.4.3, 27.7.1, 28.5.5])**.**

(1) The map

$$\begin{aligned} F\colon\ B_+^\times\backslash\widehat{B}^\times/\widehat{\mathcal{O}}^\times &\longrightarrow \operatorname{Cls}\mathcal{O}\\ B_+^\times\hat{\alpha}\widehat{\mathcal{O}}^\times &\longmapsto \hat{\alpha}\widehat{\mathcal{O}}\cap B\end{aligned}$$

is a bijection. In particular, $Y^B(\mathcal{O})$ has finitely many connected components.

(2) The map

$$\operatorname{nrd}\colon B_+^\times\backslash\widehat{B}^\times/\widehat{\mathcal{O}}^\times \longrightarrow F_+^\times\backslash\widehat{F}^\times/\operatorname{nrd}(\widehat{\mathcal{O}}^\times) \tag{14}$$

is a surjection, and if B is indefinite, then it is a bijection.

(3) When $\mathcal{O}$ is a maximal or Eichler order, $\operatorname{nrd}(\widehat{\mathcal{O}}^\times) = \widehat{\mathbb{Z}_F}^\times$ and $F_+^\times\backslash\widehat{F}^\times/\operatorname{nrd}(\widehat{\mathcal{O}}^\times)\cong\mathrm{Cl}_F^+$.

Let H be the cardinality of $B_+^\times\backslash\widehat{B}^\times/\widehat{\mathcal{O}}^\times$, and pick representatives $\{\hat{\alpha}_h\}_{h\in[H]}$. Given $\phi\in M_k^B(\mathcal{O},\Xi)$ and $h\in[H]$, we can define a function

$$\begin{aligned}\phi_h\colon\ \mathcal{H}^r &\longrightarrow W_k\\ z&\longmapsto \phi(z,\hat{\alpha}_h).\end{aligned} \tag{15}$$

For any function $\phi_h\colon\mathcal{H}^r\to W_k$, we define an action

$$\phi_h|_k\gamma := \frac{(\det\gamma)^{k_{\mathbb{R}}/2}}{j(\gamma,z)^{k_{\mathbb{R}}}}\phi_h(\gamma z)^{\gamma}. \tag{16}$$

Let $\mathcal{O}_h := \hat{\alpha}_h\widehat{\mathcal{O}}\hat{\alpha}_h^{-1}\cap B$, and define

$$M_k^B(\mathcal{O},\Xi;h) := \{\phi_h\colon\mathcal{H}^r\longrightarrow W_k\colon\ \phi_h|_k\gamma = \Xi(\hat{\alpha}_h^{-1}\gamma\hat{\alpha}_h)\phi_h\ \text{ for }\gamma\in\mathcal{O}_h^\times\}. \tag{17}$$

Note that $\hat{\alpha}_h^{-1}\gamma\hat{\alpha}_h \in \widehat{\mathcal{O}}^\times$, so it makes sense to evaluate Ξ on it.

Lemma 5.2. *The map*

$$\begin{aligned} \Phi\colon\ M_k^B(\mathcal{O},\Xi) &\longrightarrow \textstyle\bigoplus_{h=1}^H M_k^B(\mathcal{O},\Xi;h) \\ \phi &\longmapsto (\phi_h)_{h\in[H]} \end{aligned}$$

is an isomorphism.

Proof. The map Φ is well-defined because for $\gamma \in \mathcal{O}_h^\times$,

$$(\phi_h|_k\gamma)(z) \ = \frac{(\det\gamma)^{k_{\mathbb{R}}/2}}{j(\gamma,z)^{k_{\mathbb{R}}}}\phi(\gamma z,\hat{\alpha}_h)^\gamma \ = \phi(z,\gamma^{-1}\hat{\alpha}_h) \ = \phi(z,\hat{\alpha}_h(\hat{\alpha}_h^{-1}\gamma^{-1}\hat{\alpha}_h)) \ = \Xi(\hat{\alpha}_h^{-1}\gamma\hat{\alpha}_h)\phi_h(z).$$

To see that it is an isomorphism, we construct the inverse map. Given $(\phi_h)_{h\in[H]}$, define

$$\phi(z,\hat{\gamma}\hat{\alpha}_h\hat{u}) := \Xi(\hat{u})^{-1}(\phi_h|_k\gamma)(z).$$

This is well-defined because if $\hat{\gamma}\hat{\alpha}_h\hat{u} = \hat{\gamma}'\hat{\alpha}_h\hat{u}'$, then $(\gamma')^{-1}\gamma = \hat{\alpha_h}\hat{u}'\hat{u}^{-1}\hat{\alpha_h}^{-1} \in \mathcal{O}_h$ and Equation (17) gives us what we want. □

Recall the definition of Hecke operators on $M_k^B(\mathcal{O},\Xi)$ in Equation (8). We can ask how the Hecke operator interacts with the isomorphism of Lemma 5.2.

Lemma 5.3. *There exist:*

(1) A function $j^*\colon[H]\to[H]$ for every $j\in[P]$;
(2) Elements

$$\left\{\varpi_{j,h} \in \hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times\hat{\pi}_j\hat{\alpha}_h^{-1} \ \cap B_+^\times\right\}_{\substack{j\in[P]\\ h\in[H]}},$$

where each $\varpi_{j,h}$ is well-defined up to multiplication on the left by $\mathcal{O}_{j^*(h)}^\times$;

such that

$$(T_{\mathfrak{p}}\phi)_h(z) = \sum_{j=1}^P \Xi(\hat{\alpha}_h^{-1}\varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)})(\phi_{j^*(h)}|_k\varpi_{j,h})(z). \tag{18}$$

Proof. By strong approximation (Theorem 5.1), for all $h\in[H]$ and $j\in[P]$, there exist $\varpi_{j,h}\in B_+^\times$, $j^*(h)\in[H]$, and $\hat{u}\in\widehat{\mathcal{O}}^\times$ such that $\hat{\alpha}_h\hat{\pi}_j^{-1} = \varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}\hat{u}$. Applying this to Equation (8),

$$\begin{aligned}(T_{\mathfrak{p}}\phi)_h(z) &= \sum_{j=1}^P \phi(z,\varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}\hat{u})\Xi(\hat{\pi}_j)^{-1} \\ &= \sum_{j=1}^P \Xi(\hat{u}\hat{\pi}_j)^{-1}(\phi_{j^*(h)}|_k\varpi_{j,h})(z) \\ &= \sum_{j=1}^P \Xi(\hat{\alpha}_h^{-1}\varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)})(\phi_{j^*(h)}|_k\varpi_{j,h})(z). \end{aligned} \tag{19}$$

If for some j and h we have $\varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}\hat{u} = (\varpi'_{j,h})^{-1}\hat{\alpha}_{j^*(h)}\hat{u}'$, then $\varpi'_{j,h} = \gamma\varpi_{j,h}$ for some $\gamma\in\mathcal{O}_{j^*(h)}^\times$. By Equation (17), replacing $\varpi_{j,h}$ with $\varpi'_{j,h} = \gamma\varpi_{j,h}$ does not change the summand in Equation (19). □

Here is one way to think about the sets $\{\varpi_{j,h}\}_{j\in[P]}$ for each $h\in[H]$. By definition,

$$\varpi_{j,h} \in (\hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times\hat{\alpha}_{j^*(h)}^{-1})\backslash\hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times\hat{\pi}_j\hat{\alpha}_h^{-1}.$$

Applying Equation (5), we see that for a fixed h, the elements $\{\alpha_{j^*(h)}^{-1}\varpi_{j,h}\}_{j\in[P]}$ are in

$$\widehat{\mathcal{O}}^\times\backslash\widehat{\mathcal{O}}^\times\hat{\pi}\widehat{\mathcal{O}}^\times\hat{\alpha}_h^{-1} = \sqcup_{j=1}^P\widehat{\mathcal{O}}^\times\hat{\pi}_j\hat{\alpha}_h^{-1}.$$

By definition of the $\{\hat{\alpha}_h\}$ and Theorem 5.1, for each $j \in [P]$, there is a unique $\hat{\alpha}_{j^*(h)}$ for which $\hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times \hat{\pi}_j \hat{\alpha}_h^{-1} \cap B_+^\times \neq \emptyset$. Said differently, we can construct $\{\varpi_{j,h}\}_{j\in[P]}$ by considering the quotients

$$(\hat{\alpha}_{h'}\widehat{\mathcal{O}}^\times \hat{\alpha}_{h'}^{-1})\backslash \hat{\alpha}_{h'}\widehat{\mathcal{O}}^\times \hat{\pi}\widehat{\mathcal{O}}^\times \hat{\alpha}_h^{-1} = \sqcup_{j=1}^P (\hat{\alpha}_{h'}\widehat{\mathcal{O}}^\times \hat{\alpha}_{h'}^{-1})\backslash \hat{\alpha}_{h'}\widehat{\mathcal{O}}^\times \hat{\pi}_j \hat{\alpha}_h^{-1}$$

for all $h' \in [H]$. For each $j \in [P]$, there will be a unique $h' =: j^*(h)$ for which the corresponding coset has nonempty intersection with $B_+^\times$, and we take $\varpi_{j,h}$ to be any $B_+^\times$ representative of this (unique) coset.

Remark. If $H = 1$, then the quotient of interest becomes $\widehat{\mathcal{O}}^\times\backslash\widehat{\mathcal{O}}^\times \hat{\pi}\widehat{\mathcal{O}}^\times$, and we are back in the setting of Equation (6) – in this case, we can always choose $\hat{\pi}_j^{-1} = \varpi_{j,h} \in B_+^\times$, and the machinery of Lemma 5.3 is unnecessary. The complication is entirely to deal with the fact that the quaternionic Shimura variety can have many different components.

Lemma 5.4. *Fix $h \in [H]$. The set of cosets $\{\mathcal{O}_{j^*(h)}^\times \varpi_{j,h}\}_{j\in[P]}$ is invariant under multiplication on the right by $\gamma \in \mathcal{O}_h^\times$. Furthermore, if $\gamma(j) \in [P]$ is such that $\varpi_{j,h}\gamma \in \mathcal{O}_{\gamma(j)(h)}^\times \varpi_{\gamma(j),h}$, then $\gamma(j)(h) = j^*(h)$.*

Proof. Write $\gamma = \hat{\alpha}_h \hat{u}_\gamma \hat{\alpha}_h^{-1}$ for $\hat{u}_\gamma \in \widehat{\mathcal{O}}^\times$. Then, $\varpi_{j,h}\gamma \in \hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times \hat{\pi}_j^{-1}\hat{u}_\gamma\hat{\alpha}_h^{-1}$. Multiplication by an element of $\widehat{\mathcal{O}}^\times$ permutes the cosets $\sqcup\widehat{\mathcal{O}}^\times \hat{\pi}_j^{-1}$, so the result follows. □

Similarly, there exist a permutation $\sigma_\mathfrak{p} : [H] \to [H]$ and elements

$$\left\{\varpi_h \in \hat{\alpha}_{\sigma_\mathfrak{p}(h)}\widehat{\mathcal{O}}^\times \hat{\varpi}\hat{\alpha}_h^{-1} \cap B_+^\times\right\}_{h\in[H]},$$

where each ϖ_h is well-defined up to multiplication on the left by $\mathcal{O}_{\sigma_\mathfrak{p}(h)}^\times$, such that

$$(\langle\mathfrak{p}\rangle\phi)_h(z) = \Xi\left(\hat{\alpha}_h^{-1}\varpi_h^{-1}\hat{\alpha}_{\sigma_\mathfrak{p}(h)}\right)(\phi_{\sigma_\mathfrak{p}(h)}|_k\varpi_h)(z). \tag{20}$$

Again, for any h, there is a unique $\hat{\alpha}_{\sigma_\mathfrak{p}(h)}$ for which $\hat{\alpha}_{\sigma_\mathfrak{p}(h)}\widehat{\mathcal{O}}^\times \hat{\varpi}\hat{\alpha}_h^{-1} \cap B_+^\times \neq \emptyset$, and we take ϖ_h to be any representative of the corresponding coset.

[DV13, Equation 7.24] describes, in the case of trivial nebentypus, how we can avoid doing computation with adèles by replacing the double coset representatives $\{\hat{\alpha}_h\}_{h\in[H]}$ with the corresponding right $\mathcal{O}$-ideals $\{I_h := \hat{\alpha}_h\hat{\mathcal{O}} \cap B\}$ (representatives of the corresponding right ideal classes in $\mathcal{O}$) and computing the $\varpi_{j,h}$ in terms of these ideals. Observe that by Lemma 5.4, for any given $h \in [H]$, the cosets $\{\mathcal{O}_{j^*(h)}^\times\varpi_{j,h}\}_{j\in[P]}$, as well as $\mathcal{O}_{\sigma_\mathfrak{p}(h)}^\times\varpi_h$, depend only on the cosets $\{\hat{\alpha}_h\widehat{\mathcal{O}}^\times\}$, and hence only on the ideals I_h.

We would like to evaluate the expression $\Xi(\hat{\alpha}_h^{-1}\varpi_{j,h}\hat{\alpha}_{j^*(h)})$ in Equation (18) without working adèlically.

Lemma 5.5. *We can choose representatives $\{\hat{\alpha}_h\}_{h\in[H]}$ such that for all $h \in [H]$, $(\hat{\alpha}_h)_\mathfrak{p} \in \mathcal{O}_\mathfrak{p}^\times$ for all $\mathfrak{p} \mid \mathfrak{N}\Delta_B$ and $(\hat{\alpha}_h)_\mathfrak{p} \in \mathcal{O}_\mathfrak{p} \cap B_\mathfrak{p}^\times$ for all $\mathfrak{p}$.*

Proof. Pick any $\hat{\alpha} \in \widehat{B}^\times$. Because $\prod_{\mathfrak{p}\mid\mathfrak{N}\Delta_B}\mathcal{O}_\mathfrak{p}^\times$ is open in $\prod_{\mathfrak{p}\mid\mathfrak{N}\Delta_B}B_\mathfrak{p}^\times$, weak approximation [Voi21, Proposition 28.7.3] implies that there is a $\gamma \in B^\times$ such that $(\gamma\hat{\alpha})_\mathfrak{p} \in \mathcal{O}_\mathfrak{p}^\times$ for all $\mathfrak{p} \mid \mathfrak{N}\Delta_B$. Set $\hat{\alpha}' := \gamma\hat{\alpha}$. By definition of the restricted product, $\hat{\alpha}_\mathfrak{p} \in \mathcal{O}_0(1)_\mathfrak{p}^\times$ for all but finitely many $\mathfrak{p} \nmid \mathfrak{N}\Delta_B$. For $\mathfrak{p} \nmid \mathfrak{N}\Delta_B$, $\mathcal{O}_\mathfrak{p} = \mathcal{O}_0(1)_\mathfrak{p} \cong M_2(\mathbb{Z}_\mathfrak{p})$. Suppose that $\hat{\alpha}_\mathfrak{p} \in M_2(F_\mathfrak{p}) \setminus M_2(\mathbb{Z}_\mathfrak{p})$. Let $\gamma' \in F^\times$ be a totally positive generator of the principal ideal $\mathfrak{p}^{\#\mathrm{Cl}_F^+}$. Viewed as an element of $B^\times$, $\gamma'_\mathfrak{q} = 1$ for all primes $\mathfrak{q} \neq \mathfrak{p}$, and For some r, $\gamma_\mathfrak{p}^r\hat{\alpha}'_\mathfrak{p} \in M_2(\mathbb{Z}_\mathfrak{p})$. Therefore, we can replace $\hat{\alpha}'$ by $\gamma^r\hat{\alpha}'$. Repeating this for every prime where $\hat{\alpha}'$ is not integral, we obtain a $\hat{\alpha}' \in B^\times\hat{\alpha}\widehat{\mathcal{O}}^\times$ satisfying the conditions of the lemma. □

For $\{\hat{\alpha}_h\}$ as in Lemma 5.5, $\varpi_{j,h} \in \hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times\hat{\pi}_j\hat{\alpha}_h^{-1}$ lies in $\mathcal{O}_\mathfrak{p}$ for every $\mathfrak{p} \mid \mathfrak{N}$, and we have

$$\Xi(\hat{\alpha}_h^{-1}\varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}) = \Xi(\hat{\alpha}_h^{-1})\Xi(\varpi_{j,h}^{-1})\Xi(\hat{\alpha}_{j^*(h)}).$$

Similarly, $\varpi_h \in \hat{\alpha}_{\sigma_\mathfrak{p}(h)}\widehat{\mathcal{O}}^\times\hat{\varpi}\hat{\alpha}_h^{-1}$ lies in $\mathcal{O}_\mathfrak{p}$ for every $\mathfrak{p} \mid \mathfrak{N}$, and we have

$$\Xi(\hat{\alpha}_h^{-1}\varpi_h^{-1}\hat{\alpha}_{\sigma_\mathfrak{p}(h)}) = \Xi(\hat{\alpha}_h^{-1})\Xi(\varpi_h^{-1})\Xi(\hat{\alpha}_{\sigma_\mathfrak{p}(h)}).$$

On the level of ideals, the $\{\hat{\alpha}_h\}$ being as in Lemma 5.5 is equivalent to the ideals $\{I_h\}$ being integral and coprime to $\mathfrak{N}$. Let $T_\mathfrak{p}$ be the matrix of the Hecke operator on $\bigoplus_h M_k^B(\mathcal{O}, \Xi; h)$ defined by Equation (18), and let $T'_{\mathfrak{p}}$ be the matrix defined by Equation (18) after replacing $\Xi(\hat{\alpha}_h^{-1}\varpi_h^{-1}\hat{\alpha}_{j^*(h)})$ with $\Xi(\varpi_{j,h}^{-1})$. Let

$$D := \operatorname{diag}\left(\dots, \underbrace{\Xi(\hat{\alpha}_h)^{-1}, \dots, \Xi(\hat{\alpha}_h)^{-1}}_{\dim M_k^B(\mathcal{O},\Xi;h) \text{ times}}, \dots\right).$$

Then, $T_{\mathfrak{p}} = DT'_{\mathfrak{p}}D^{-1}$. Since D is independent of $\mathfrak{p}$, the $\{T'_{\mathfrak{p}}\}$ and $\{T_{\mathfrak{p}}\}$ differ only by a change of basis. Therefore, we lose nothing by computing $\{T'_{\mathfrak{p}}\}$ – avoiding any adelic computation – once we have chosen $\{\hat{\alpha}_h\}$ as in Lemma 5.5.

Remark. If Ξ is finitely ramified, we can even choose representatives $\{\hat{\alpha}_h\}_h \subset \ker \Xi$ – in the proof of Lemma 5.5, we can add the primes at which Ξ is ramified to the set which we control using weak approximation so that $(\hat{\alpha}_h)_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}^{\times}$, and then multiply on the right by an element of $\widehat{\mathcal{O}}^{\times}$ that makes $(\hat{\alpha}_h)_{\mathfrak{p}} = 1$ for all these primes. However, – setting the nebentypus factor aside – our algorithms involve the ideals $\{I_h\}$ rather than the adèlic elements $\{\hat{\alpha}_h\}$ – while we can still require that I_h be integral and coprime to the conductor of Ξ, the condition that $\Xi(\hat{\alpha}_h) = 1$ is not detected by the coset $\hat{\alpha}_h\widehat{\mathcal{O}}^{\times}$ and hence cannot be expressed in terms of the $\{I_h\}$.

5.2. **Why Lemma 5.3 is not enough.** Lemma 5.3 is not quite computationally useful. When $r = 1$, the spaces $M_k^B(\mathcal{O}, \Xi; h)$ in Equation (17) are spaces of holomorphic functions on $\mathcal{H}$ satisfying conditions similar to those defining a modular form. The Hecke action in Lemma 5.3 generalizes the classical formula

$$T_p f = \sum_j f|_k \omega_j,$$

for the Hecke operator on a modular form. This is not ideal for computational purposes – we would like a description involving only discrete quantities and linear algebra, with no analog objects in the mix.

When $r = 0$, the quaternionic Shimura variety is a disjoint union of points, and each $M_k(\mathcal{O}, \Xi; h)$ can be specified by an element of W_k. Then, Lemma 5.3 actually gives us a usable formula for computing the Hecke action on this space. However, this procedure is inefficient when computing spaces of many different levels $\mathcal{O}$ (e.g. when computing tables of Hilbert modular forms) as for each $\mathcal{O}$ we need new sets of representatives $\{\hat{\alpha}_h\}$ and elements $\{\varpi_{j,h}\}$ and $\{\varpi_h\}$.

In the following sections, we rectify these issues. In particular, we describe how to compute spaces with varying $\mathcal{O} = \mathcal{O}_0(\mathfrak{N})$ using only the $\{\hat{\alpha}_h\}$ and $\{\varpi_{j,h}\}$, $\{\varpi_h\}$ associated to $\mathcal{O} = \mathcal{O}_0(1)$ (or slight modifications thereof).

6. Indefinite method

In this section, we will describe a procedure for computing $S_k^B(\mathcal{O}, \Xi)$ when $r = 1$, i.e. when B is split at exactly one infinite place. Algorithms to compute $M_k^B(\mathcal{O}, 1)$ for such B were developed by [GV11]. They were extended to fields with arbitrary narrow class number by [Voi10]. In this section, we explain how to extend their methods to arbitrary nebentypus Ξ. The idea of these approaches is to use a generalization of the Eichler-Shimura isomorphism (Theorem 6.3) to realize each $S_k^B(\mathcal{O}, \Xi; h)$ from Lemma 5.2 within the first cohomology of an arithmetic group (depending on the level $\mathcal{O}$ and on h) with some coefficients (depending on the weight k and nebentypus Ξ).

We start with an example of the method applied to weight 2 modular forms with nebentypus. Before we begin, we will first need to extend the action of $\mathrm{PSL}_2(\mathbb{R})$ on $\mathcal{H}$ to an action of $\mathrm{PGL}_2(\mathbb{R})$. Define, for $\mu = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R}) \setminus \mathrm{SL}_2(\mathbb{R})$ and $z \in \mathcal{H}$,

$$\mu z := \frac{a\bar{z} + b}{c\bar{z} + d}.$$

Example 6.1. Let $B = M_{2,\mathbb{Q}}$, $\mathcal{O} = \mathcal{O}_0(N) \subseteq M_2(\mathbb{Z})$ an order, and χ a semigroup homomorphism of $\mathcal{O}$ associated to a finite order character of $\mathbb{Z}/N\mathbb{Z}$ as in Example 3.1. Let $\tilde{\Gamma} := \mathcal{O}^{\times}/\pm 1 \subset \mathrm{PGL}_2(\mathbb{Z})$ and $\Gamma := \mathcal{O}^1/\pm 1 \subset \mathrm{PSL}_2(\mathbb{Z})$, where $\mathcal{O}^1$ denotes the elements of $\mathcal{O}^{\times}$ with determinant 1. First, given $f \in S_2(\Gamma, \chi)$, the differential $\omega_f := f(z)dz$ defines a closed holomorphic differential on $\mathcal{H}$. It is not quite Γ-invariant because χ is nontrivial, but writing $Y := \Gamma\backslash\mathcal{H}$ and X for its compactification, one can check that $\gamma^*\omega_f = \chi(\gamma)\omega_f$. Because f is a cusp form, $\omega_f \in H^0(X, \mathbb{C}(\chi) \otimes_{\underline{\mathbb{C}}_X} \Omega_X^1)$, where $\mathbb{C}(\chi)$ is the local system associated to the representation χ of $\Gamma = \pi_1(X) \subset \mathcal{O}$. The map $f \mapsto \omega_f$ induces an injection (and in fact an isomorphism, by Lemma 6.5) from $S_2(\Gamma, \chi)$ to $H^0(X, \mathbb{C}(\chi) \otimes_{\mathbb{C}_X} \Omega^1_X)$.

Given a base point $P \in X$, the integral

$$\begin{aligned} \xi_f\colon\ \Gamma &\longrightarrow \mathbb{C}(\chi) \\ \gamma &\longmapsto \textstyle\int_P^{\gamma P} \omega_f(z) \end{aligned}$$

is well-defined (it depends only on the homotopy class of the path $P \sim \gamma P$ in $\mathcal{H}$). The map ξ_f is in fact a 1-cocycle, as

$$\int_P^{\gamma\gamma' P} \omega_f \ = \xi_f(\gamma) + \int_P^{\gamma' P} \gamma^* \omega_f \ = \xi_f(\gamma) + \chi_0(\gamma) \int_P^{\gamma' P} \omega_f \ = \xi_f(\gamma) + \gamma \cdot \xi_f(\gamma').$$

Composing the map taking f to ω_f with the map taking ω_f to ξ_f, we obtain a map

$$\Phi\colon S_2(\Gamma, \chi) \xrightarrow{\sim} H^0(X, \mathbb{C}(\chi) \otimes_{\underline{C}_X} \Omega^1_X) \hookrightarrow H^1(\Gamma, \mathbb{C}(\chi)).$$

By Theorem 6.9, this is actually an injection of Hecke modules.

Take $\mu = \begin{pmatrix} -1 & \\ & 1 \end{pmatrix} \in \tilde{\Gamma} \backslash \Gamma$ – because $\tilde{\Gamma}/\Gamma \cong \mathbb{Z}/2\mathbb{Z}$, any choice of μ will work, but this choice is convenient. Then,

$$\int_P^{\mu\gamma\mu^{-1} P} f(z) dz = \int_{\mu P}^{\mu\gamma P} f(z) dz = \int_P^{\gamma P} f(\mu z) d(\mu z) = \int_P^{\gamma P} -f(-\overline{z}) d\overline{z}.$$

As such, $\mu \bullet \xi_f$ is antiholomorphic. We can check that if $f(z) = \sum_m a_m \exp(2\pi\sqrt{-1} m z)$, then $\overline{f(-\overline{z})} = \sum_m \overline{a_m} \exp(2\pi\sqrt{-1} m z) =: f^c(z)$. In particular, it follows (again from Theorem 6.9) that the action of μ commutes with the Hecke action, and so we have a $\mathbb{C}$-linear injection of Hecke module maps

$$S_2(\Gamma, \chi) \xrightarrow{\Phi} H^1(\Gamma, \mathbb{C}(\chi)) \xrightarrow{\xi \longmapsto (\xi + \mu \bullet \xi)} H^1(\Gamma, \mathbb{C}(\chi))^{\tilde{\Gamma}/\Gamma}. \tag{21}$$

In the main result of this section, Theorem 6.3, we replace Γ with an arbitrary arithmetic Fuchsian group over a totally real field, and allow higher weight. Nonetheless, the idea of the proof is exactly the same as that of Example 6.1. In fact, because we assume that $B \ncong M_{2,F}$, we get a Hecke module isomorphism instead of just the injection of Equation (21).

6.1. **Defining the coefficients of the cohomology.** The representation $\operatorname{Sym}^m \mathbb{C}^2$ can be described as the space of homogeneous bivariate degree m polynomials $\mathbb{C}[\mathsf{X}, \mathsf{Y}]_m$ (which we think of as functions on the column vector $(\mathsf{X}\ \mathsf{Y})^T$) with a right action of $\mathrm{GL}_2(\mathbb{C})$ given by precomposing with $\gamma \in \mathrm{GL}_2(\mathbb{C})$. We will instead want to think of this as a space with a left[2] action of $\gamma \in \mathrm{GL}_2(\mathbb{C})$ given by precomposing with the transpose γ^T, i.e. $\gamma \cdot f(\mathsf{X}, \mathsf{Y}) = f(a\mathsf{X} + b\mathsf{Y}, c\mathsf{X} + d\mathsf{Y})$.

Let k_1 denote the component of the weight at the (unique) infinite place of F split in B. Then,

$$V_k := \left((\operatorname{Sym}^{k_1 - 2} \mathbb{C}^2) \otimes \det^{2 - k_1/2} \right) \otimes W_k \cong \bigotimes_{i \in [n]} \operatorname{Sym}^{k_i - 2} \mathbb{C}^2 \otimes \det^{(\underline{k_0} - k_i)/2}, \tag{22}$$

where the isomorphism follows from the self-duality of $(\operatorname{Sym}^{k_i - 2} \mathbb{C}^2) \otimes \det^{2 - k_i/2}$. We equip this space with a left action of $B^\times$ whereby given $\gamma \in B^\times$ and $v \otimes w \in \mathbb{C}[\mathsf{X}, \mathsf{Y}]_{k_1} \otimes W_k$, we define

$$\gamma \cdot (v \otimes w) = (\gamma \cdot v) \otimes w^{\gamma^{-1}} = v \circ \gamma^T \otimes w^{\gamma^{-1}}.$$

We define $\Xi_h\colon \widehat{\mathcal{O}}_h \cap B^\times \to \mathbb{C}$ by $\Xi_h(\hat{x}) := \Xi(\hat{\alpha}_h^{-1} \hat{x} \hat{\alpha}_h)$, and set $V_{k, \Xi_h} := V_k \otimes \Xi_h$.

Remark 6.2. *While V_k is a representation of $B^\times$, $V_{k,\Xi}$ is only a representation of of $\widehat{\mathcal{O}}_h^\times$. When defining the Hecke action in Equation (29), we will need to act on V_k by the elements $\{\varpi_{j,h}\}$ from Lemma 5.3, which are not in $\mathcal{O}_h^\times$ for any given $h \in [H]$. This is why we notate the V_k action (through the operator $\cdot$) and the nebentypus separately.*

[2]The convention in the literature is to give V_k a right action. However, I felt that this would make the proof of Theorem 6.3 less readable.

By the Hasse-Schilling theorem (see [Voi21, Main Theorem 14.7.4]) (or simply by noting that the reduced norm on $\mathbb{H}^\times$ is always positive), for any $\gamma \in B^\times$, $\mathrm{nrd}(\gamma)$ is positive at every infinite place of F which is ramified in B. Since $r = 1$, this means that $\mathrm{nrd}(\gamma)$ is positive at all but one infinite place. Let $\mathcal{O}_h^1 := \{\gamma \in \mathcal{O}_h^\times : \mathrm{nrd}(\gamma) = 1\}$, $\tilde{\Gamma}_h := \mathcal{O}_h^\times/(\mathbb{Z}_F^\times \cap \mathcal{O}_h^\times)$, and $\Gamma_h := \mathcal{O}_h^1/(\mathbb{Z}_F^\times \cap \mathcal{O}_h^1)$. Because B is indefinite, $\mathcal{O}_h^\times$ contains an element of negative (in the embedding corresponding to the split infinite place) norm by [Voi21, Theorem 28.6.1]. Thus, there is an exact sequence

$$1 \longrightarrow \Gamma_h \longrightarrow \tilde{\Gamma}_h \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 1. \tag{23}$$

Because V_{k,Ξ_h} is trivial on scalars, it descends to a representation of $B^\times/F^\times$.

6.2. **The Eichler-Shimura isomorphism.** With this in hand, we now prove the generalization of Equation (21) that we need.

6.3. **Proving a vector space isomorphism.**

Theorem 6.3. *Let B/F be split at exactly one infinite place. Then, there is a $\mathbb{C}$-linear isomorphism*

$$S_k^B(\mathcal{O}, \Xi; h) \cong H^1(\Gamma_h, V_{k,\Xi_h})^{\tilde{\Gamma}_h/\Gamma_h}, \tag{24}$$

and in particular

$$S_k^B(\mathcal{O}, \Xi) \cong \bigoplus_{h \in [h^+(F)]} H^1(\Gamma_h, V_{k,\Xi_h})^{\tilde{\Gamma}_h/\Gamma_h}.$$

For the remainder of this subsection, we prove Theorem 6.3. Because we prove the isomorphism of Equation (24) for each $h \in [H]$ separately, we may simplify our notation by writing $\Xi := \Xi_h$, $\tilde{\Gamma} := \tilde{\Gamma}_h$, $\Gamma := \Gamma_h$, and $X := X_h := \Gamma_h \backslash \mathcal{H}$.

Given $f \in S_k(\mathcal{O}, \Xi; h)$, we associate to f a $V_{k,\Xi} = \mathbb{C}[\mathsf{X}, \mathsf{Y}]_{(k_1-2)} \otimes W_k \otimes \Xi$-valued differential 1-form on $\mathcal{H}$:

$$\omega_f(z) := f(z)(z\mathsf{X} + \mathsf{Y})^{k-2} dz.$$

This differential is holomorphic and closed. Rather than directly showing it is Γ-invariant (and hence that it descends to $X(\Gamma)$, we prove a slightly more general fact which will be useful later when studying the Hecke action. Given $\beta \in B^\times$ and a $V_{k,\Xi}$-valued differential form ω on $\mathcal{H}$, we write $\beta \cdot \omega$ for the form produced by postcomposing ω with the left V_k-action of β – as discussed in Remark 6.2, we explicitly do not include the nebentypus twist in the $\cdot$ notation.

Lemma 6.4. *Let $\beta \in B_+^\times$. Then,*

$$\beta^* \omega_{(f|_k \beta^{-1})} = \beta \cdot \omega_f.$$

Proof. For $\beta = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we have

$$\begin{aligned} \omega_{f|_k \beta^{-1}}(\beta z) &= (\det \beta)^{2-k_1/2}(cz+d)^{k_1-2} f(z)(\frac{az+b}{cz+d}\mathsf{X} + \mathsf{Y})^{k_1-2} dz \\ &= (\det \beta)^{2-k_1/2} f(z)((a\mathsf{X} + c\mathsf{Y})z + (b\mathsf{X} + d\mathsf{Y}))^{k_1-2} dz \\ &= \beta \cdot \omega_f. \end{aligned}$$

□

It follows from Equation (17) and Lemma 6.4 that for $\gamma \in \Gamma$,

$$\gamma^* \omega_f = \gamma^* \omega_{\Xi(\gamma) f|_k \gamma^{-1}} = \Xi(\gamma)\gamma \cdot \omega_f. \tag{25}$$

In particular, ω_f descends to a $V_{k,\Xi}$-valued differential form on X. Equation (25) also implies that

$$\int_{\gamma A}^{\gamma B} \omega_f = \Xi(\gamma)\gamma \cdot \int_A^B \omega_f. \tag{26}$$

We denote by $\mathcal{V}_{k,\Xi}$ the local system on X corresponding to the monodromy representation $V_{k,\Xi}$, and write $(\mathcal{V}_{k,\Xi}, \nabla)$ (where $\mathcal{V}_{k,\Xi} := V_{k,\Xi} \otimes_{\mathbb{C}_X} \mathcal{O}_X$) for the (holomorphic) vector bundle with flat connection whose flat sections are $V_{k,\Xi}$. Then, $\omega_f \in H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1)$ by definition.

Lemma 6.5. *The map sending f to ω_f induces a $\mathbb{C}$-linear isomorphism from $S_k^B(\mathcal{O}, \Xi; h)$ to $H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1)$.*

Sketch. The form ω_f is zero if and only if f is. For surjectivity, first observe that any $\omega \in A^{1,0}(\mathcal{H}, V_{k,\Xi})$ (a smooth $V_{k,\Xi}$-valued $(1,0)$-form on the upper half-plane $\mathcal{H}$) can be written as $P(\mathsf{X},\mathsf{Y};z)dz$ on $\mathcal{H}$, where $P(\mathsf{X},\mathsf{Y};z)$ is a degree (k_1-2)-homogeneous polynomial in X and Y whose coefficients are smooth functions of z valued in $W_k \otimes \Xi$. We claim that any $\omega \in H^0(\mathcal{H}, \mathcal{V}_{k,\Xi} \otimes \Omega^1) \subset A^{1,0}(\mathcal{H}, V_{k,\Xi})$ (i.e. a holomorphic $V_{k,\Xi}$-valued $(1,0)$-form) arising as the pullback of a form from $X(\Gamma)$ can be written as $f(z)(z\mathsf{X}+\mathsf{Y})^{k-2}$ for some $f \in S_k^B(\mathcal{O}, \Xi; h)$.

For brevity, let $m := k_1 - 2$. Making a linear change of variables $\mathsf{Z} := z\mathsf{X}+\mathsf{Y}$, $\overline{\mathsf{Z}} := \overline{z}\mathsf{X}+\mathsf{Y}$, we can write

$$P(\mathsf{X},\mathsf{Y};z) = \sum_{i=0}^{m} f_i(z)\mathsf{Z}^i\overline{\mathsf{Z}}^j$$

for smooth functions $f_i\colon \mathcal{H} \to \mathbb{C}$. The form ω is holomorphic if and only if $\overline{\partial}\omega = 0$ We compute

$$\begin{aligned}\frac{\partial}{\partial \overline{z}} f_i(z)\mathsf{Z}^i\overline{\mathsf{Z}}^{m-i} &= \frac{\partial f_i}{\partial \overline{z}}\mathsf{Z}^i\overline{\mathsf{Z}}^{m-i} + f_i(z)\mathsf{Z}^i((m-i)\mathsf{X}\overline{\mathsf{Z}}^{m-i-1}) \\ &= \frac{\partial f_i}{\partial \overline{z}}\mathsf{Z}^i\overline{\mathsf{Z}}^{m-i} + (m-i)f_i(z)\frac{\mathsf{Z}+\overline{\mathsf{Z}}}{z-\overline{z}}\mathsf{Z}^i\overline{\mathsf{Z}}^{m-1-i} \\ &= \left(\frac{\partial f_i}{\partial \overline{z}} + \frac{(m-i)f_i(z)}{z-\overline{z}}\right)\mathsf{Z}^i\overline{\mathsf{Z}}^{m-i} + \frac{(m-i)f_i(z)}{z-\overline{z}}\mathsf{Z}^{i+1}\overline{\mathsf{Z}}^{m-i-1}.\end{aligned}$$

Therefore, writing $f_{-1} = 0$,

$$\overline{\partial}\omega = \sum_{i=0}^{k-2}\left(\frac{\partial f}{\partial \overline{z}} + \frac{(m-i)f_i(z)}{z-\overline{z}} + \frac{(m-i-1)f_{i-1}(z)}{z-\overline{z}}\right)\mathsf{Z}^i\overline{\mathsf{Z}}^{m-i},$$

and for ω to be holomorphic all of these terms must vanish. At $i=0$, the condition is that

$$\frac{\partial f_0}{\partial \overline{z}} = \frac{k_1-2}{z-\overline{z}} f_0.$$

Solving this differential equation, we find that $f_0(z) = \frac{g_0(z)}{(z-\overline{z})^{2-k_1}}$ for some holomorphic function $g_0(z)$. Because we want ω to descend to a well defined form on the compact quotient $\Gamma\backslash\mathcal{H}$, we cannot allow $f_0(z)$ to diverge as $\mathrm{Im}(z) \to \infty$, and therefore we must have $g_0(z) = f_0(z) = 0$. We then repeat this for $i=1$ and so on, until we reach $i = k_1 - 2$. Then, the condition becomes $\frac{\partial f_m}{\partial \overline{z}} = 0$, and hence that $f_m(z)$ is holomorphic. Therefore, using only holomorphicity and "boundedness as $z \to \infty$", we have shown that $\omega = f(z)(z\mathsf{X}+\mathsf{Y})^{k_1-2}dz$ for some holomorphic function f valued in $W_k \otimes \Xi$. In order for ω to descend to a form on $X(\Gamma)$, we also have the condition $\omega = \gamma^{-1}\cdot\gamma^*\omega$ for any $\gamma \in \Gamma$, or equivalently,

$$\begin{aligned}f(z)(z\mathsf{X}+\mathsf{Y})^{k_1-2}dz &= f\left(\frac{az+b}{cz+d}\right)\left(\frac{az+b}{cz+d}(d\mathsf{X}-c\mathsf{Y}) + (-b\mathsf{X}+a\mathsf{Y})\right)\frac{dz}{(cz+d)^2} \\ &= f\left(\frac{az+b}{cz+d}\right)(z\mathsf{X}+\mathsf{Y})^{k_1-2}dz\end{aligned}$$

for all $\gamma \in \Gamma$. Therefore, $\omega = f(z)(z\mathsf{X}+\mathsf{Y})^{k-2}dz$, where $f \in S_k^B(\mathcal{O}, \Xi; h)$ as desired. $\square$

Lemma 6.6.

$$H^1(\Gamma, V_{k,\Xi}) \cong S_k^B(\mathcal{O}, \Xi; h) \oplus \overline{S_k^B(\mathcal{O}, \Xi^{-1}; h)}. \tag{27}$$

This should look like the usual statement of the Eichler-Shimura isomorphism with nebentypus [Hid93, Chapter 6]. The proof of Lemma 6.6 uses Hodge theory, and relies on the compactness of $X = \Gamma\backslash\mathcal{H}$ (because $B \not\cong M_{2,F}$). Otherwise, we would have contributions coming from Hilbert Eisenstein series.

Proof. Building the Hodge-de Rham spectral sequence from the resolution

$$0 \longrightarrow V_{k,\Xi} \hookrightarrow \mathcal{V}_k \xrightarrow{\nabla} \mathcal{V}_k \otimes \Omega^1 \xrightarrow{\nabla} \mathcal{V}_k \otimes \Omega^2 \xrightarrow{\nabla} \cdots$$

of $V_{k,\Xi}$, we obtain a Hodge decomposition

$$H^1(\Gamma, V_{k,\Xi}) \cong H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1) \oplus H^1(X, \mathcal{V}_{k,\Xi} \otimes \mathcal{O}_X).$$

By Serre duality,
$$H^1(X, \mathcal{V}_{k,\Xi} \otimes \mathcal{O}_X) \cong H^0(X, \mathcal{V}_{k,\Xi}^* \otimes \Omega_X^1)^*.$$
Since $\mathcal{V}_{k,\Xi}$ is the vector bundle associated to the representation $V_{k,\Xi}$, one can check that $\mathcal{V}_{k,\Xi}^*$ is the vector bundle associated to the dual representation $V_{k,\Xi}^* \cong V_{k,\Xi^{-1}}$.

By the Hodge index theorem, which goes through in the same way with local system coefficients, there is a Hermitian pairing on $H^1(\Gamma_h, V_{k,\Xi}^*)$ which restricts to a Hermitian pairing on $H^0(X, \mathcal{V}_{k,\Xi}^* \otimes \Omega_X^1)$. We deduce
$$H^0(X, \mathcal{V}_{k,\Xi}^* \otimes \Omega_X^1)^* \cong \overline{H^0(X, \mathcal{V}_{k,\Xi}^* \otimes \Omega_X^1)} \cong \overline{H^0(X, \mathcal{V}_{k,\Xi^{-1}} \otimes \Omega_X^1)}.$$
The result then follows from Lemma 6.5. □

Pick an arbitrary basepoint $P \in \mathcal{H}$. To any $f \in S_k^B(\mathcal{O}, \Xi; h)$, we can associate a function
$$\begin{aligned} \xi_f \colon\ \Gamma_h &\longrightarrow W_k \\ \gamma &\longmapsto \int_{P_h}^{\gamma P_h} f(z)(z\mathsf{X} + \mathsf{Y})^{k_1 - 2} dz \end{aligned}$$
The function ξ_f is a 1-cocycle, as
$$\int_P^{\gamma\gamma' P} \omega_f = \int_P^{\gamma P} \omega_f + \int_{\gamma P}^{\gamma\gamma' P} \omega_f \ = \xi_f(\gamma) + \gamma \cdot \xi_f(\gamma').$$
Because ω_f is closed, this integral only depends on the path from P to γP up to homotopy. The class of ξ_f in $H^1(\Gamma, V_k)$ is independent of the choice of base point P. As such, we have a well-defined composition $F\colon S_k^B(\mathcal{O}, \Xi) \xrightarrow{f \mapsto \omega_f} H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1) \to H^1(\Gamma, V_{k,\Xi})$ sending f to ξ_f. The first map is an isomorphism by Lemma 6.5, and the second is an injection because any (($V_{k,\Xi}$)-valued) differential form on $X(\Gamma)$ is determined by its (($V_{k,\Xi}$)-valued)-integrals on a homology basis for $X(\Gamma)$.

In what follows, let $Z := \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}$. (This is distinct from the variable Z used in the proof of Lemma 6.5.)

Lemma 6.7. *For $f \in S_k^B(\mathcal{O}, \Xi; h)$ and $\mu \in \tilde{\Gamma} \setminus \Gamma$,*
$$\mu^{-1} \cdot (\mu^* \omega_f) = \overline{\omega_{f^c}}$$
for $f^c := f(-\overline{z})^Z \in S_k^B(\mathcal{O}, \Xi^{-1}; h)$.

Proof.
$$\begin{aligned} \mu^{-1} \cdot (\mu^* \omega_f(z)) &= \mu^{-1} \cdot \left(f(\mu z) \left(\frac{a\overline{z} + b}{c\overline{z} + d} \mathsf{X} + \mathsf{Y} \right)^{k_1 - 2} d(\mu z) \right) \\ &= f((\mu Z^{-1}) Z z)^{\mu} (c\overline{z} + d)^{2 - k_1} (\overline{z}\mathsf{X} + \mathsf{Y})^{k_1 - 2} d(\mu z) \\ &= f(-\overline{z})^Z (\overline{z}\mathsf{X} + \mathsf{Y})^{k_1 - 2} d\overline{z}. \end{aligned}$$
This is certainly an antiholomorphic differential. We want to show that $f^c(z) := f(-\overline{z})^Z$ is in $S_k^B(\mathcal{O}, \Xi^{-1}; h)$. For $\gamma \in \Gamma_h$,
$$(f^c|_k\gamma)(z) = \frac{(\det\gamma)^{k_1/2}}{(cz+d)^{k_1}} \left(\overline{f((Z\gamma Z^{-1})Zz)^Z} \right)^{\gamma} \ = \frac{(\det\gamma)^{k_1/2}}{(cz+d)^{k_1}} \left(\overline{\Xi(\gamma) \frac{(c\overline{z}+d)^{k_1}}{(\det\gamma)^{k_1/2}} f(Zz)^{Z\gamma^{-1}}} \right)^{\gamma} \ = \Xi^{-1}(\gamma) f^c(z).$$
□

Corollary 6.8. *For $f \in S_k^B(\mathcal{O}, \Xi; h)$ and $\mu \in \tilde{\Gamma} \setminus \Gamma$,*
$$\mu \bullet \xi_f = \overline{\xi_{f^c}}.$$

Proof. We have
$$\mu^{-1} \cdot \int_P^{\mu\gamma\mu^{-1} P} \omega_f \ = \int_{\mu^{-1} P}^{\gamma\mu^{-1} P} \mu^{-1} \cdot \mu^* \omega_f \ = \int_{\mu^{-1} P}^{\gamma\mu^{-1} P} \overline{\omega_{f^c}}.$$
Because changing the basepoint from $\mu^{-1} P$ to P does not affect the cohomology class of ξ_g, the result follows. □

It follows that the μ action on $H^1(X, V_{k,\Xi})$ interchanges the pieces of the Hodge decomposition in Equation (27). Writing $H^1(X, V_{k,\Xi})^{\mu=1}$ for the fixed points under the μ-action, the dimension of $H^1(X, V_{k,\Xi})^{\mu=1}$ is exactly half that of $H^1(X, V_{k,\Xi})$, and in particular

$$\dim S_k^B(\mathcal{O}, \Xi; h) = \dim H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1) \; = \frac{1}{2} \dim H^1(X, V_{k,\Xi}) = \dim H^1(\Gamma_h, V_{k,\Xi})^{\mu=1}.$$

We can realize the isomorphism by the map

$$S_k^B(\mathcal{O}, \Xi) \xrightarrow{f \longmapsto \omega_f} H^0(X, \mathcal{V}_{k,\Xi} \otimes \Omega_X^1) \xrightarrow{\omega_f \longmapsto \xi_f} H^1(\Gamma, V_{k,\Xi}) \xrightarrow{\xi_f \longmapsto (\xi_f + \mu \bullet \xi_f)} H^1(\Gamma, V_{k,\Xi})^{\mu=1} \tag{28}$$

This concludes the proof of Theorem 6.3.

6.4. **Proving a Hecke module isomorphism.** We now proceed to describe the Hecke action on $\bigoplus_h H^1(\Gamma_h, V_{k,\Xi_h})$. The Hecke operators will interlace the data from the various direct summands, so our notation will need to keep track of this – as such, we reintroduce the subscripts $h \in [H]$ going forward. We write $\mathcal{H}_h$ for the h^{th} copy of the upper half plane. In this language, $Y^B(\mathcal{O}) = \sqcup_h \Gamma_h \backslash \mathcal{H}_h$. Our objects of study will be an automorphic form $\phi = (\phi_h)_{h \in [H]} \in \bigoplus_h S_k^B(\mathcal{O}, \Xi; h) \cong S_k^B(\mathcal{O}, \Xi)$, the associated differential forms $\omega = (\omega_{\phi_h})_{h \in [H]}$, where ω_{ϕ_h} is a differential form on $\Gamma_h \backslash \mathcal{H}_h$, and the associated cocycles $\xi = (\xi_{\phi_h})_{h \in [H]} \in \bigoplus_h H^1(\Gamma_h, V_{k,\Xi_h})$. With this notation in hand, we now translate the Hecke action given in Lemma 5.3 into cohomology. Define

$$(T_{\mathfrak{p}} \xi)_h(\gamma) := \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_{\gamma(j),h}^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j),h}^{-1} \cdot \xi_{j^*(h)}(\varpi_{\gamma(j),h} \gamma \varpi_{j,h}^{-1}). \tag{29}$$

Theorem 6.9. *For $\xi = (\xi_h)_h$ and $\phi = (\phi_h)_h$ as above,*

$$\xi_{(T_{\mathfrak{p}} \phi)_h} = (T_{\mathfrak{p}} \xi)_h \in H^1(\Gamma_h, V_{k,\Xi_h})$$

for all $h \in [H]$.

Proof. For brevity, we write $\varpi_j := \varpi_{j,h}$. Let $(P_h)_{h \in [H]}$ be a collection of (arbitrary) basepoints $P_h \in \mathcal{H}_h$. By Lemma 5.4 that for any $\gamma \in \mathcal{O}_j^\times$, there is a unique $\gamma(j) \in [P]$ for which $\varpi_{\gamma(j)} \gamma \varpi_j^{-1} \in B^\times$, and for this $\gamma(j)$, $\gamma(j)(h) = j^*(h)$.

In what follows, we think of the action of ϖ_j on $\sqcup \mathcal{H}_h$ as sending $\mathcal{H}_h$ to $\mathcal{H}_{j^*(h)}$ – this is important so that we only integrate differentials on $\mathcal{H}_h$ against paths on $\mathcal{H}_h$. As discussed below Lemma 5.4, we may assume without loss of generality that $\Xi(\hat{\alpha}_h) = 1$ for all h and that $\varpi_{j,h}$ is locally in $\mathcal{O}_{\mathfrak{p}} \cap B_{\mathfrak{p}}^\times$ for every $\mathfrak{p} \nmid \mathfrak{N}$. For $\gamma \in \Gamma_h$,

$$\begin{aligned}
\xi_{(T_{\mathfrak{p}} \phi)_h}(\gamma) &= \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_j^{-1} \hat{\alpha}_{j^*(h)}) \int_{P_h}^{\gamma P_h} \omega_{\phi_{j^*(h)}|_k \varpi_{j,h}} \\
&= \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_j^{-1} \hat{\alpha}_{j^*(h)}) \varpi_j^{-1} \cdot \int_{\varpi_j P_h}^{\varpi_j \gamma P_h} \omega_{\phi_{j^*(h)}} \\
&= \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_j^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j)}^{-1} \cdot \int_{\varpi_{\gamma(j)} P_h}^{\varpi_{\gamma(j)} \gamma P_h} \omega_{\phi_{j^*(h)}},
\end{aligned}$$

where the first equality is Equation (18), the second applies Lemma 6.4, and the third uses that $\gamma(j)(h) = j^*(h)$. We can rewrite this as

$$\sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_{\gamma(j)}^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j)}^{-1} \cdot \left(\int_{P_{j^*(h)}}^{\varpi_{\gamma(j)} \gamma \varpi_j^{-1} P_{j^*(h)}} \omega_{\phi_{j^*(h)}} \right) + \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_{\gamma(j)}^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j)}^{-1} \cdot \left(\int_{\varpi_{\gamma(j)} \gamma \varpi_j^{-1} P_{j^*(h)}}^{\varpi_{\gamma(j)} \gamma P_h} \omega_{\phi_{j^*(h)}} + \int_{\varpi_{\gamma(j)} P_h}^{P_{j^*(h)}} \omega \right. \tag{30}$$

The first summand in Equation (30) is exactly $(T_{\mathfrak{p}} \xi)_h$ by Equation (29), so it remains to show that

$$\sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_{\gamma(j)}^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j)}^{-1} \cdot \left(\int_{\varpi_{\gamma(j)} \gamma \varpi_j^{-1} P_{j^*(h)}}^{\varpi_{\gamma(j)} \gamma P_h} \omega_{\phi_{j^*(h)}} \right) + \sum_j \Xi(\hat{\alpha}_h^{-1} \varpi_{\gamma(j)}^{-1} \hat{\alpha}_{j^*(h)}) \varpi_{\gamma(j)}^{-1} \cdot \left(\int_{\varpi_{\gamma(j)} P_h}^{P_{j^*(h)}} \omega_{\phi_{j^*(h)}} \right). \tag{31}$$

is a 1-coboundary. Let $\gamma' := \varpi_{\gamma(j)}\gamma\varpi_j^{-1}$. Substituting this in and applying Equation (26), the first term in Equation (31) becomes

$$\sum_j \Xi(\hat{\alpha}_h^{-1}\varpi_{\gamma(j)}^{-1}\hat{\alpha}_{j^*(h)})\varpi_{\gamma(j)}^{-1} \cdot \int_{\gamma' P_{j^*(h)}}^{\gamma'\varpi_j P_h} \omega_{\phi_{j^*(h)}} = \sum_j \Xi(\hat{\alpha}_h^{-1}\varpi_{\gamma(j)}^{-1}\gamma'\hat{\alpha}_{j^*(h)})(\varpi_{\gamma(j)}^{-1}\gamma') \cdot \int_{P_{j^*(h)}}^{\varpi_j P_h} \omega_{\phi_{j^*(h)}}. \tag{32}$$

Reindexing and noting again that $\gamma(j)(h) = j^*(h)$, the second term in Equation (31) is

$$\sum_j \Xi(\hat{\alpha}_h^{-1}\varpi_j^{-1}\hat{\alpha}_{j^*(h)})\varpi_j^{-1} \cdot \int_{\varpi_j P_h}^{P_{j^*(h)}} \omega_{\phi_{j^*(h)}}. \tag{33}$$

The sum of Equation (32) and Equation (33) is then

$$\begin{aligned}
&\sum_j \left(\Xi(\hat{\alpha}_h^{-1}\varpi_{\gamma(j)}^{-1}\gamma'\hat{\alpha}_{j^*(h)})(\varpi_{\gamma(j)}^{-1}\gamma') - \Xi(\hat{\alpha}_h^{-1}\varpi_j^{-1}\hat{\alpha}_{j^*(h)})\varpi_j^{-1}\right) \cdot \int_{P_{j^*(h)}}^{\varpi_j P_h} \omega_{\phi_{j^*(h)}} \\
=&\sum_j (\Xi_h(\gamma^{-1})\gamma - 1) \cdot \left(\Xi(\hat{\alpha}_{j^*(h)}^{-1}\varpi_j\hat{\alpha}_h)\varpi_j^{-1} \cdot \int_{P_{j^*(h)}}^{\varpi_j P_h} \omega_{\phi_{j^*(h)}}\right).
\end{aligned}$$

This is a 1-coboundary, so we are done. □

Similarly, we can define

$$(\langle \mathfrak{p} \rangle \xi)_h(\gamma) := \Xi(\hat{\alpha}_h^{-1}\varpi_h^{-1}\hat{\alpha}_{\sigma_{\mathfrak{p}}(h)})\varpi_h^{-1} \cdot \xi_{\sigma_{\mathfrak{p}}(h)}(\varpi_h\gamma\varpi_h^{-1}), \tag{34}$$

and then we have $\xi_{(\langle \mathfrak{p} \rangle \phi)_h} = (\langle \mathfrak{p} \rangle \xi)_h$

Let $\mu = (\mu_h)_{h\in[H]}$ be a collection of $\mu_h \in \tilde{\Gamma}_h/\Gamma_h$. As noted in Corollary 6.8, the specific choices of μ_h do not matter. We let μ act on $\xi = (\xi_h)_h$ componentwise.

Lemma 6.10. *The actions of μ and $T_{\mathfrak{p}}$ on $\bigoplus_h H^1(\Gamma_h, V_{k,\Xi_h})$ commute.*

Proof. By Lemma 5.4, for all $j \in [P]$ there is a permutation $\mu_h \colon [P] \to [P]$ such that $\varpi_{j,h}\mu_h = \mu'_{j^*(h)}\varpi_{\mu(j),h}$, where $\mu'_{j^*(h)} \in \tilde{\Gamma}_{j^*(h)} \setminus \Gamma_{j^*(h)}$ since $\{\varpi_{j,h}\}_j \subset B_+^\times$. Noting that $\mu(j)(h) = j^*(h)$ by Lemma 5.4,

$$\begin{aligned}
(T_{\mathfrak{p}}(\mu \bullet \xi))_h &= \sum_j \Xi(\hat{\alpha}_h^{-1}\varpi_{\gamma(j),h}^{-1}\hat{\alpha}_{j^*(h)})(\varpi_{\gamma(j)}^{-1}\mu_{j^*(h)}^{-1}) \cdot \xi_{j^*(h)}(\mu_{j^*(h)}\varpi_{\gamma(j)}\gamma\varpi_j^{-1}\mu_{j^*(h)}) \\
&= \sum_j \Xi(\hat{\alpha}_h^{-1}\varpi_{\gamma(j),h}^{-1}\hat{\alpha}_{j^*(h)})((\mu'_h)^{-1}\varpi_{\mu_{j^*(h)}(\gamma(j))}^{-1}) \cdot \xi_{j^*(h)}(\varpi_{\mu_{j^*(h)}(\gamma(j))}\mu'_h\gamma(\mu'_h)^{-1}\varpi_{\mu_{j^*(h)}(j)}^{-1}) \\
&= (\mu \bullet (T_{\mathfrak{p}}\xi))_h,
\end{aligned}$$

where in the last equality we use that the choices of $\{\mu_h\}_h$ do not affect the action of μ on cohomology. □

Similarly, these actions commute with the action of $\langle \mathfrak{p} \rangle$. We can thus compute the actions of $T_{\mathfrak{p}}$ and $\langle \mathfrak{p} \rangle$ on $S_k^B(\mathcal{O}, \Xi)$ cohomologically, by computing the action on $\bigoplus_h H^1(\Gamma_h, V_{k,\Xi_h})$ using Equation (29) and Equation (34) and then projecting to the subspace where μ acts trivially. The actual computation of the cohomology groups is essentially exactly as described in [DV21, Voi10, DV13]. Let $\Gamma_h(1) := \mathcal{O}_0(1)^1/\mathbb{Z}_F^\times$. By Shapiro's lemma,

$$H^1(\Gamma_h, V_{k,\Xi_h}) \cong H^1\left(\Gamma_1, \operatorname{Ind}_{\Gamma_h}^{\Gamma_h(1)} V_{k,\Xi_h}\right).$$

By [?, Lemma 1.1.4], the Hecke action commutes with Shapiro's lemma for $\mathfrak{p} \nmid \mathfrak{N}$, meaning that by working with the induced representation we can compute Hecke operators for such $\mathfrak{p}$ at level $\mathcal{O}$ while only using the $\{\varpi_{j,h}\}$ of $\mathcal{O}_0(1)$. When $\mathfrak{p} \mid \mathfrak{N}$, we can still compute Hecke operators, but we need to first undo the Shapiro isomorphism, compute the Hecke operator on the level $\mathcal{O}$ space, and then reapply the Shapiro isomorphism. Despite this, we can still get away with using more or less the original $\varpi_{j,h}$, since if $\hat{\alpha}_h\pi_j^{-1} \in \varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}_0(1)}^\times$, then by multiplying $\varpi_{j,h}^{-1}$ on the left by an element of $\mathcal{O}_{j^*(h)}^\times$, we can find an element of $\mathcal{O}_{j^*(h)}^\times\varpi_{j,h}$ for which $\hat{\alpha}_h\pi_j^{-1} \in \varpi_{j,h}^{-1}\hat{\alpha}_{j^*(h)}\widehat{\mathcal{O}}^\times$. Similarly, we can still use the original ϖ_h.

7. Definite method

In this section, we describe a procedure for computing $S_k^B(\mathcal{O},\Xi)$ when B is (totally) definite, i.e. when $r=0$. Definition 4.1 simplifies in this case, and $M_k^B(\mathcal{O},\Xi)$ consists of functions $\phi\colon \widehat{B}^\times \to W_k$ such that $\phi(\gamma\hat{x})^\gamma = \phi(\hat{x})$ and $\phi(\hat{x}\hat{u}) = \Xi(\hat{u})\varphi(\hat{x})$. As discussed in Section 5.2, Lemma 5.3 already gives us a way to compute the Hecke action on this space, but requires us to compute a new set of elements $\{\varpi_{j,h}\}$ at each level $\mathcal{O}$ we are considering.

In this section, assuming that $\mathcal{O}=\mathcal{O}_0(\mathfrak{N})$ is an Eichler order of some level $\mathfrak{N}$, we bypass this issue by doing precomputation at level $\mathcal{O}_0(1)$ and working with the induced representation $\operatorname{Ind}_{\widehat{\mathcal{O}}^\times}^{\widehat{\mathcal{O}_0(1)}^\times}\Xi$ This is superficially similar to how we used Shapiro's lemma at the end of Section 7, but note that here the induction is taking place in the domain of the functions in question rather than in the codomain. In this section we bypass Section 4 and start from Definition 4.1 anew. We do not directly use Lemma 5.3, though the $\{\varpi_{j,h}\}$ which appeared there will again appear here.

Such algorithms to compute $M_k^B(\mathcal{O},1)$ for definite B were developed by [Dem07] and extended to fields with arbitrary narrow class number by [DD08]. In this section, we explain how to extend their methods to arbitrary nebentypus Ξ. A procedure for doing so was sketched at the end of [Dem07, Section 4] – the purpose of this section is to spell out all of the details. Note that our induced representations are exactly the same as Dembélé's twisted projective lines.

Let $H_1 := \#\operatorname{Cls}\mathcal{O}_0(1)$ and $H := \#\operatorname{Cls}\mathcal{O}$. By Lemma 5.2, $M_k(\mathcal{O},\Xi)\cong\bigoplus_{h'\in[H]} M_k(\mathcal{O},\Xi;h')$. Let $\{\hat{\alpha}_h\}_{h\in[H_1]}$ be double coset representatives for $B^\times\backslash\widehat{B}^\times/\widehat{\mathcal{O}_0(1)}^\times$.

Because $\mathcal{O}=\mathcal{O}_0(\mathfrak{N})$ and $(\mathfrak{N},\Delta_B)=1$, there are isomorphisms $\widehat{\mathcal{O}_0(1)}_h^\times/\widehat{\mathcal{O}}_h^\times \cong \widehat{\mathcal{O}_0(1)}^\times/\widehat{\mathcal{O}}^\times \cong \mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})$. Similarly, $\operatorname{Ind}_{\widehat{\mathcal{O}}_h^\times}^{\widehat{\mathcal{O}_0(1)}_h^\times}\Xi_h \cong \operatorname{Ind}_{\widehat{\mathcal{O}}^\times}^{\widehat{\mathcal{O}_0(1)}^\times}\Xi$. Given representatives $\{\hat{\kappa}_x\}_{x\in\mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})}$ for $\widehat{\mathcal{O}_0(1)}^\times/\widehat{\mathcal{O}}^\times$, we have representatives $\{\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}\}_{x\in\mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})}$ for $\widehat{\mathcal{O}_0(1)}_h^\times/\widehat{\mathcal{O}}_h^\times$. We can take the vectors $[\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}]_{x\in\mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})}$ to be a basis for $\operatorname{Ind}_{\widehat{\mathcal{O}}_h^\times}^{\widehat{\mathcal{O}_0(1)}_h^\times}\Xi_h$. For any $\beta\in\widehat{\mathcal{O}_0(1)}_h^\times$ and any coset representative $\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}$, there is a unique $\beta(x)\in\mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})$ such that $\beta\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}\in(\hat{\alpha}_h\hat{\kappa}_{\beta(x)}\hat{\alpha}_h^{-1})\widehat{\mathcal{O}}_h^\times$. Equivalently, defining $\kappa_\beta := \hat{\alpha}_h^{-1}\beta\hat{\alpha}_h$, $\kappa_{\beta(x)}^{-1}\kappa_\beta\kappa_x\in\widehat{\mathcal{O}}^\times$. We obtain an explicit realization of $\operatorname{Ind}_{\widehat{\mathcal{O}}_h^\times}^{\widehat{\mathcal{O}_0(1)}_h^\times}\Xi_h$ by defining

$$\beta[\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}] := \Xi_h(\hat{\alpha}_h\hat{\kappa}_x^{-1}\hat{\kappa}_\beta^{-1}\hat{\kappa}_{\beta(x)}\hat{\alpha}_h^{-1})[\hat{\kappa}_{\beta(x)}] = \Xi(\hat{\kappa}_x^{-1}\hat{\kappa}_\beta^{-1}\hat{\kappa}_{\beta(x)})[\hat{\alpha}_h\hat{\kappa}_{\beta(x)}\hat{\alpha}_h^{-1}]. \tag{35}$$

Lemma 7.1. *For all $h\in[H_1]$, there is a $\mathbb{C}$-linear map*

$$\Phi_h\colon M_k^B(\mathcal{O},\Xi)_h \longrightarrow \{F_h\colon \operatorname{Ind}_{\widehat{\mathcal{O}}_h^\times}^{\widehat{\mathcal{O}_0(1)}_h^\times}\Xi_h \longrightarrow W_k \mid F_h(\gamma\cdot y)^\gamma = F_h(y) \textit{ for all } \gamma\in\mathcal{O}_0(1)_h^\times, y\in\widehat{\mathcal{O}_0(1)}_h^\times\}$$

$$\phi_h \longmapsto \big(F_h\colon [\hat{\alpha}_h\hat{\kappa}_y\hat{\alpha}_h^{-1}]\longmapsto \phi_h(\hat{\alpha}_h\kappa)\big)_h$$

such that

$$M_k^B(\mathcal{O},\Xi)\cong\bigoplus_{h\in[H_1]}\operatorname{im}\Phi_h.$$

Proof. We have the following sequence of bijections:

$$B^\times\backslash\hat{B}^\times/\widehat{\mathcal{O}}^\times \leftrightarrow \bigsqcup_h B^\times\backslash B^\times\hat{\alpha}_h\widehat{\mathcal{O}_0(1)}^\times/\widehat{\mathcal{O}}^\times \leftrightarrow \bigsqcup_h(\hat{\alpha}_h^{-1}B^\times\hat{\alpha}_h\cap\widehat{\mathcal{O}_0(1)}^\times)\backslash\widehat{\mathcal{O}_0(1)}^\times/\widehat{\mathcal{O}}^\times \leftrightarrow \bigsqcup_h\mathcal{O}_0(1)_h^\times\backslash\widehat{\mathcal{O}_0(1)}_h^\times/\widehat{\mathcal{O}}_h^\times,$$

with corresponding maps (skipping the leftmost)

$$\hat{\alpha}_h\hat{\kappa}\longmapsto\hat{\kappa}\longmapsto\hat{\alpha}_h\hat{\kappa}\hat{\alpha}_h^{-1}.$$

As such, we can think of left $B^\times$-equivariant functions on $\widehat{B}^\times/\widehat{\mathcal{O}}^\times$ as left $\mathcal{O}_0(1)_h^\times$-equivariant functions on $\widehat{\mathcal{O}_0(1)}_h^\times/\widehat{\mathcal{O}}_h^\times$.

For $\gamma\in\mathcal{O}_0(1)_h^\times$, let $\hat{\kappa}_\gamma := \hat{\alpha}_h^{-1}\gamma\hat{\alpha}_h\in\widehat{\mathcal{O}_0(1)}^\times$. By Equation (35),

$$F_h(\gamma\cdot[\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}])^\gamma = \Xi(\hat{\kappa}_x^{-1}\hat{\kappa}_\gamma^{-1}\hat{\kappa}_{\gamma(x)})F_h([\hat{\alpha}_h\hat{\kappa}_{\gamma(x)}\hat{\alpha}_h^{-1}])^\gamma.$$

Applying the definitions of F_h and then Definition 4.1,

$$\Xi(\hat{\kappa}_x^{-1}\hat{\kappa}_\gamma^{-1}\hat{\kappa}_{\gamma(x)})\phi_h(\gamma^{-1}\hat{\alpha}_h\hat{\kappa}_{\gamma(x)}) \;=\phi_h(\gamma^{-1}\hat{\alpha}_h\hat{\kappa}_\gamma\hat{\kappa}_x) \;=\phi_h(\hat{\alpha}_h\hat{\kappa}_x) \;=F_h([\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}])$$

The map Φ sending $\phi\in M_k^B(\mathcal{O},\Xi)$ to $(F_h)_h$ has an inverse map Ψ such that $\phi' := \Psi((F_h)_h)$ is defined by

$$\phi'(\gamma\hat{\alpha}_h\hat{\kappa}_x\hat{u}) := \Xi(\hat{u})^{-1}F_h([\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}])^{\gamma^{-1}}.$$

This is well-defined as if $\gamma\hat{\alpha}_h\hat{\kappa} = \gamma'\hat{\alpha}_h\hat{\kappa}'$, then $\gamma(\gamma')^{-1}\in\mathcal{O}_0(1)_h^\times$, and the result follows from the defining property of F_h. □

We now describe how the Hecke operators act through the isomorphism of Lemma 7.1. There is an injection of double coset spaces

$$\widehat{\mathcal{O}}^\times\backslash\widehat{\mathcal{O}}^\times\hat{\pi}\widehat{\mathcal{O}}^\times \hookrightarrow \widehat{\mathcal{O}_0(1)}^\times\backslash\widehat{\mathcal{O}_0(1)}^\times\hat{\pi}\widehat{\mathcal{O}_0(1)}^\times$$

because both quotients are trivial away from $\mathfrak{p}$. This injection is a bijection when $\mathfrak{p}\nmid\mathfrak{N}$.

In particular, if we have chosen $\{\hat{\pi}_j\}_{j\in[P]}$ such that $\widehat{\mathcal{O}}^\times\hat{\pi}\widehat{\mathcal{O}}^\times = \bigsqcup_{j\in[P]}\widehat{\mathcal{O}}^\times\hat{\pi}_j$, then $\widehat{\mathcal{O}_0(1)}^\times\hat{\pi}\widehat{\mathcal{O}_0(1)}^\times = \bigsqcup_{j\in[P]}\widehat{\mathcal{O}_0(1)}^\times\hat{\pi}_j$ if $\mathfrak{p}\nmid\mathfrak{N}$, and otherwise there is one extra coset at level $\mathcal{O}_0(1)$. Removing this extra coset if necessary, we can choose the $\{\hat{\pi}_j\}$ at level $\mathcal{O}_0(1)$ regardless of which level we ultimately want to compute at.

For $j\in[P]$ and $h\in[H_1]$, let $\{\varpi_{j,h}\}$ be as in Lemma 5.3. They satisfy

$$\hat{\alpha}_h\pi_j^{-1} = \varpi_{j,h}^{-1}\alpha_{j^*(h)}\widehat{\mathcal{O}_0(1)}^\times. \tag{36}$$

Define

$$\varpi\cdot[\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}] := \Xi(\hat{\kappa}_x^{-1}\hat{\alpha}_h^{-1}\varpi^{-1}\hat{\alpha}_{\varpi(h)}\hat{\kappa}_{\varpi(x)})[\hat{\alpha}_{j^*(h)}\hat{\kappa}_{\varpi(x)}\hat{\alpha}_{j^*(h)}^{-1}].$$

and

$$(T_\mathfrak{p}F)_h(\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}) := \sum_j F_{j^*(h)}(\varpi_{j,h}\cdot(\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}))^{\varpi_{j,h}}. \tag{37}$$

Similarly, define

$$(\langle\mathfrak{p}\rangle F)_h(\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}) := F_{\sigma_\mathfrak{p}(h)}(\varpi_h\cdot(\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}))^{\varpi_h}. \tag{38}$$

Theorem 7.2. *With respect to the Hecke actions defined in Equation (8) and Equation (37), the isomorphism of Lemma 7.1 is Hecke equivariant.*

Proof. By Equation (8), for $\phi\in M_k(\mathcal{O},\Xi)$ we have

$$\Phi(T_\mathfrak{p}\phi)_h(\hat{\alpha}_h\hat{\kappa}_x\hat{\alpha}_h^{-1}) \;=(T_\mathfrak{p}\phi)(\hat{\alpha}_h\hat{\kappa}_x) \;=\sum_j\phi(\hat{\alpha}_h\hat{\kappa}_x\pi_j^{-1})\Xi(\pi_j)^{-1}. \tag{39}$$

We cannot directly apply Equation (36) to Equation (39) because $\hat{\kappa}_x$ is in between $\hat{\alpha}_h$ and $\hat{\pi}_j^{-1}$. By Equation (5) however, because right multiplication by $\widehat{\mathcal{O}_0(1)}^\times$ permutes the cosets $\{\widehat{\mathcal{O}_0(1)}^\times\hat{\pi}_j\}$, left multiplication by $\widehat{\mathcal{O}_0(1)}^\times$ permutes the cosets $\hat{\pi}_j^{-1}\widehat{\mathcal{O}_0(1)}^\times$ and hence there exists a permutation x (abusing notation) of $[\mathrm{Nm}(\mathfrak{p})+1]$ such that $\hat{\kappa}_x\hat{\pi}_j^{-1} = \hat{\pi}_{x(j)}^{-1}\hat{\kappa}'$ for $\hat{\kappa}'\in\widehat{\mathcal{O}_0(1)}^\times$. Then,

$$\hat{\alpha}_h\hat{\kappa}_x\hat{\pi}_j^{-1} \;=\hat{\alpha}_h\hat{\pi}_{x(j)}^{-1}\hat{\kappa}' \;=\varpi_{x(j),h}^{-1}\hat{\alpha}_{x(j)(h)}\hat{\kappa}'' \;=\varpi_{x(j),h}^{-1}\hat{\alpha}_{x(j)(h)}\hat{\kappa}_y\hat{u}_{j,x,h} \tag{40}$$

for some $y := \varpi_{x(j),h}(x)\in\mathbb{P}^1(\mathbb{Z}_F/\mathfrak{N})$, $\kappa,\kappa''\in\widehat{\mathcal{O}_0(1)}^\times$, and $u_{j,x,h}\in\widehat{\mathcal{O}}^\times$. Plugging this into Equation (39), the expression becomes

$$\begin{aligned}\sum_j\phi(\varpi_{x(j),h}^{-1}\hat{\alpha}_{x(j)(h)}\hat{\kappa}_y\hat{u}_{j,x,h})\Xi(\pi_j^{-1}) &= \sum_j\phi(\hat{\alpha}_{x(j)(h)}\hat{\kappa}_y)^{\varpi_{x(j),h}}\Xi(\hat{\pi}_j^{-1}\hat{u}_{j,x,h}^{-1})\\ &= \sum_j F_{x(j)(h)}([\hat{\alpha}_{x(j)(h)}\kappa_y\hat{\alpha}_{x(j)(h)}^{-1}])^{\varpi_{x(j),h}}\Xi(\hat{\kappa}_x^{-1}\hat{\alpha}_h^{-1}\varpi_{x(j)}^{-1}\hat{\alpha}_{x(j)(h)}\hat{\kappa}_y)\\ &= \sum_j F_{j^*(h)}(\varpi_j\cdot[\hat{\alpha}_{j^*(h)}\kappa_x\hat{\alpha}_{j^*(h)}^{-1}])^{\varpi_{j,h}},\end{aligned}$$

exactly as in Equation (37). The same proof shows equivariance with respect to $\langle\mathfrak{p}\rangle$. □

With Theorem 7.2 in hand, the implementation details of computing Hecke matrices are the same as those that arise in [Dem07, DD08, DV13].

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